



YASHWANTRAO CHAVAN MAHARASHTRA OPEN UNIVERSITY
SCHOOL OF SCIENCES

(FORMERLY SCHOOL OF ARCHITECTURE, SCIENCE & TECHNOLOGY)



UGV03: B.Sc. (Honours) (Mathematics)
{2025 Pattern}
(Open Elective for Other Faculty)

NEP 2020
Compliant

MATE101:
Mathematics for
Competitive Exam

(2 Credits)
Open Elective

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VICE CHANCELLOR’S MESSAGE

Dear Students,

Greetings!!!

I offer cordial welcome to all of you for the Undergraduate Degree programme of Yashwantrao Chavan Maharashtra Open University, offered by the School of Sciences.

As an undergraduate student, you have the autonomy to learn and gain information and knowledge regarding different dimensions in the field of Mathematics. At the same time, intellectual development is necessary for application of knowledge wisely. The process of learning includes appropriate thinking, understanding key concepts, describing these concepts based on experience and observation, explaining them to others through speaking, writing or demonstrating. Today, the science of education acknowledges the principle that it is possible to achieve excellence and knowledge in this regard.

The syllabus of this course has been structured in this book in such a way, to give you autonomy to study easily without stirring from home. During the counseling sessions, scheduled at your respective learner support centre (LSC), all your doubts will be clarified about the course and you will get guidance from well qualified and experienced counselors/professors. This guidance will not only be based on lectures, but it will also include various techniques such as question-answers, doubt clarification. We expect your active participation in the contact sessions at the LSC. Our emphasis is on ‘self-study’. If a student learns how to learn, he will become independent in learning throughout life. This course book has been written with the objective of helping in self-study and giving you autonomy to learn at your convenience, provide stimulating learning experience, of course, enjoyable and at the same time prepare for gaining, fulfilling employment.

During this academic year, you have to give assignments, complete laboratory activities, field visits and the Project work wherever required. You have to opt for specialization as per programme structure. You will get experience and joy in personally doing above activities. This will enable you to assess your own progress and thereby achieve a larger educational objective.

We wish that you will enjoy the courses of Yashwantrao Chavan Maharashtra Open University, emerge successful and very soon become a knowledgeable and honorable Degree holder of this university, which is fully aligned with the features and framework of the National Education Policy (NEP) 2020.

I congratulate “Development Team” for the development of this excellent high quality “Self- Learning Material (SLM)” for the students. I hope and believe that this SLM will be immensely useful for all students of this programme.

Best Wishes!

- Prof. Sanjeev Sonawane
Vice-Chancellor, YCMOU

FOREWORD BY THE DIRECTOR

Dear Students,

Greetings!!!

This book serves the primary goal of acquainting undergraduate level students, particularly those in the field of Science, with the conceptual and practical foundations of Mathematics with applications essential for their studies. It offers a complete exploration of Mathematical principles and their practical applications. Throughout the book, you will find numerous illustrative examples designed to enhance your understanding and skills while cultivating a deep appreciation for the applications.

We have adopted a "learn at your own pace" approach, making each unit of the book as user-friendly as possible. To do this, the book is written in a self-instructional format, ensuring that each unit follows a carefully structured approach. This not only facilitates comprehension but also keeps the learning process engaging.

Each unit of book commences with clearly defined learning objectives, using action verbs aligned with Bloom's Taxonomy to help you understand what you will achieve. Additionally, each unit starts with an introduction to stimulate the learner's curiosity and enthusiasm for the upcoming content. As you progress, you'll encounter detailed explanations supported by tables, figures, exhibits, and solved illustrations, all thoughtfully included enhancing your ability how to learn.

This Self-Learning Material (SLM) book is written in simple language with conversational style, concise sentences and covers the entire syllabus. The topics within each unit are presented in a logical sequence, starting from simple concepts and progressing to more complex one, ensuring that the material is approachable for learners with varying levels of intellectual capacity.

To enrich learning experiences, each unit contains a diverse range of exercises, including multiple-choice questions, conceptual questions, and practical problems. These exercises aim to guide students in understanding every aspect of a particular concept, gradually building their knowledge and skills.

We are grateful to the learners, resourceful writers, diligent editors, and the dedicated School faculty and staff who contributed to the development of this SLM book.

Hope this book becomes a valuable companion to support your academic pursuits and enrich your understanding of content covered in this course. We are excited to support you on this rewarding journey, in line with the holistic and multidisciplinary vision of NEP 2020.

Let's Grow Together....Best Wishes to all of you!!

- Dr. Chetana Kamlaskar

Director,

School of Sciences, YCMOU

LEARNING OBJECTIVES AND COURSE OUTCOMES

LEARNING OBJECTIVES:

The Learning objectives of this course are -

- To strengthen fundamental numerical skills required for competitive examinations.
- To enable students to compute H.C.F., L.C.M., square roots, and cube roots using efficient methods.
- To develop proficiency in simplification techniques for fast and accurate calculation.
- To build conceptual understanding of ratios, proportions, percentages, and logarithms.
- To introduce sequences and series for solving progression-based problems.

COURSE OUTCOMES:

After successful completion of this course, student should be able to –

- Apply basic number properties and operations to solve exam-oriented numerical problems.
- Calculate H.C.F., L.C.M., square roots, cube roots, and perform simplifications using shortcut techniques.
- Demonstrate competency in solving questions based on ratios, proportions, percentages, and logarithms.
- Identify and compute terms of common sequences and series used in competitive exams.
- Apply learned mathematical techniques to improve speed and accuracy in competitive examination problem-solving.

CREDIT 01

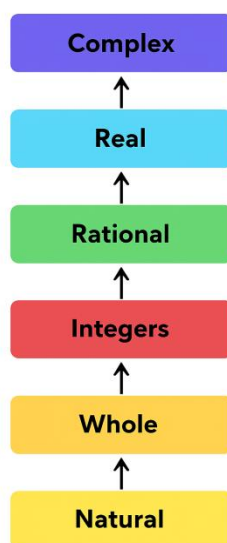
UNIT 01-01: NUMBERS

LEARNING OBJECTIVES

After successful completion of this unit, you will be able to

- ❖ Identify and classify different types of numbers in the number system.
- ❖ Distinguish between natural, whole, integers, rational, irrational, real, and complex numbers.
- ❖ Understand and apply the properties of numbers under basic operations.
- ❖ Differentiate between prime and composite numbers.

INTRODUCTION



When we think of mathematics, the first thing that comes to mind is numbers. Typically, we envision the digits 0, 1, 2, and so on. However, these are not the only numbers. Numbers form the foundation of mathematics, and there are various types that are essential for solving problems in competitive exams. A clear understanding of number systems, their classifications, and properties enables quick and accurate problem-solving. This unit will discuss different types of numbers, including prime and composite numbers, along with their operations and properties.

01-01-01: NUMBERS & TYPES OF NUMBERS

In decimal number system, we have ten digits, namely 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.

A group of digits denote a number.

Digits in a group denoting a number have two types of values face value and place value.

The face value of a digit in a number is its own value; at whatever place it may be.

To find the place value of a digit in a number following place-value chart is used:

Place	Ten Crores	Crores	Ten Lakhs	Lakhs	Ten Thousands	Thousands	Hundreds	Tens	Ones
Place Value	Digit x 100000000	Digit x 10000000	Digit x 1000000	Digit x 100000	Digit x 10000	Digit x 1000	Digit x 100	Digit x 10	Digit x 1

Illustration: Consider, a number **73619548**.

In this given number 73619548 face and place value of each digit is:

Number	7	3	6	1	9	5	4	8
Face Value	7	3	6	1	9	5	4	8
Place Value	7 x 100000000	3 x 10000000	6 x 100000	1 x 10000	9 x 1000	5 x 100	4 x 10	8 x 1

Illustration: Write a number 95636147 in words.

To write a number in words place values are used.

Place	Ten Crores	Crores	Ten Lakhs	Lakhs	Ten Thousands	Thousands	Hundreds	Tens	Ones
Number		9	5	6	3	6	1	4	7

95636147 in words can be written as **Nine crore fifty-six lakhs thirty-six thousand one hundred forty-seven.**

Illustration: Write the number Twenty-one crore sixty lakh five thousand two hundred fourteen in figures:

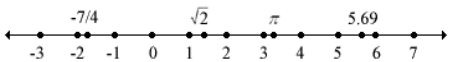
Place	Ten Crores	Crores	Ten Lakhs	Lakhs	Ten Thousands	Thousands	Hundreds	Tens	Ones
Number	2	1	6	0	0	5	2	1	4

Given number in figure: **216005214.**

Types of Numbers

SN	Type	Definition	Example
1.	Natural Numbers	Counting numbers	{1, 2, 3, ...}
2.	Whole Numbers	All counting numbers, together with 0	• {0, 1, 2, 3, ...} • All natural numbers are whole numbers • 0 is a whole number which is not a natural number
3.	Integers	All counting numbers, zero and negatives of counting numbers	{..., -3, -2, -1, 0, 1, 2, 3, ...}
			Positive Integers = {1, 2, 3, 4, 5, 6, ... }
			Negative Integers = { -1, -2, -3, -4, -5, -6, ... }
			Non-negative Integers = {0, 1, 2, 3, 4, 5, ... }
			Non-positive Integers = { ..., -5, -4, -3, -2, -1, 0 }
6.	Even Numbers	Integer divisible by 2 is called an even number	{..., -6, -4, -2, 0, 2, 4, 6, ...}
7.	Odd Numbers	Integer not divisible by 2 is called an odd number	{..., -7, -5, -3, -1, 1, 3, 5, 7, ...}
			• $\text{Odd} \pm \text{Odd} = \text{Even}$ • $\text{Even} \pm \text{Even} = \text{Even}$ • $\text{Odd} \pm \text{Even} = \text{Odd}$ • $\text{Even} \pm \text{Odd} = \text{Odd}$ • $(\text{Even})^{\text{Odd}} = \text{Even}$ • $(\text{Odd})^{\text{Even}} = \text{Odd}$

8.	Prime Numbers	A counting number which is divisible by 1 and itself only is known as a prime number	Prime numbers between 1 and 100: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97 • The smallest prime number is 2. • The only even prime number is 2. • The first odd prime number is 3. • 1 is a unique number – neither prime nor composite.
9.	Composite Numbers	All counting numbers, which are not prime	4,6,8,9 • Lowest composite number = 4. • The least odd composite number is 9.
10.	Co-primes (or Relative Primes)	Two numbers whose H.C.F. is 1 are called co-prime numbers	(2, 3), (8, 9) are pairs of co-primes.
11.	Twin Primes	Two prime numbers whose difference is 2 are called twin-primes	(3, 5), (5, 7), (11, 13) are pairs of twin-primes.
12.	Rational Number	A number which can be expressed as a fraction $\frac{p}{q}$ where p and q are integers and $q \neq 0$	$-\frac{3}{8}, \frac{4}{7}, \frac{23}{6}$

13	Irrational Numbers	- Any real number that is not rational - Numbers which when expressed in decimal would be in non-terminating and non-repeating form	$\sqrt{2}, \sqrt{3}, e, 0.231764735\dots$
14	Real number	- Every real number is represented by a unique point on the number line. - Union of the rational and irrational numbers	 - Real line on the left hand side extends to $-\infty$ and to $+\infty$ on the right hand side - On the number line, numbers on the right hand side are always larger than the numbers on the left hand side
15	Imaginary number	Imaginary numbers cannot be plotted on the number line	$i, 2i, -5i$ where $i = \sqrt{-1}$

Note: Natural Numbers in a range

If a range of numbers is given then natural numbers in that range can be calculated as given below:

1. $A = [15, 45] = 15 \leq N \leq 45$, here both the ends of range are included.

$$\text{Natural numbers in } A = (45 - 15) + 1 = 31$$

2. $B = [15, 45) = 15 \leq N < 45$, here only one end 15 of range is included.

$$\text{Natural numbers in } B = (45 - 15) = 30$$

3. $C = (15, 45] = 15 < N \leq 45$, here only one end 45 of range is included.

$$\text{Natural numbers in } C = (45 - 15) = 30$$

4. $D = (15, 45) = 15 < N < 45$, here none of the ends of range are included.

$$\text{Natural numbers in } C = (45 - 15) - 1 = 29$$

Special Number Forms:

For any natural number n ,

1. $n(n + 1)$ is always even.

Illustration: $7 \times 8 = 56, 10 \times 11 = 110$ is even.

2. $n(n + 1)(n + 2)$ is always multiple of 6.

Illustration: $5 \times 6 \times 7 = 210$ is multiple of 6.

3. If n is odd then $n^2 - 1$ is always divisible by 8.

Illustration: $5^2 - 1 = 25 - 1 = 24$ is divisible by 8

4. $n^3 - n$ is always divisible by 6.

Illustration: $4^3 - 4 = 64 - 4 = 60$ is divisible by 6.

5. $n^2 + n$ is always even.

Illustration: $7^2 + 7 = 49 + 7 = 56$ is even.

Important Number System Formulae:

1. Sum of first n natural numbers $= \frac{n(n+1)}{2}$

Illustration: Sum of first 15 natural numbers $= 1 + 2 + 3 + \dots + 15 =$
 $\frac{15(15+1)}{2} = 120$

2. Sum of Squares of first n natural numbers $= \frac{n(n+1)(2n+1)}{6}$

Illustration: Sum of Squares of first 15 natural numbers $= 1 + 2^2 + 3^2 + \dots +$
 $15^2 = \frac{15(15+1)(2 \times 15 + 1)}{6} = \frac{15 \times 16 \times 31}{6} = 1240$

3. Sum of Cubes of first n natural numbers $= \left[\frac{n(n+1)}{2} \right]^2$

Illustration: Sum of Cubes of first 15 natural numbers $= 1 + 2^3 + 3^3 + \dots +$
 $15^3 = \left[\frac{15(15+1)}{2} \right]^2 = (120)^2 = 14400$

4. Sum of first n even natural numbers $= n(n + 1)$

Illustration: Sum of first 10 even natural numbers $= 2 + 4 + 6 + 8 + 10 + 12 + 14 + 16 + 18 + 20 = 10(10 + 1) = 110$

5. Sum of first n odd natural numbers $= n^2$

Illustration: Sum of first 10 odd natural numbers $= 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 = 10^2 = 100$

SOLVED PROBLEMS

1) **Problem 01-01-01:** What is the sum of the greatest and smallest number of five digits?

Solution: Greatest number of five digits $= 99999$

Smallest number of five digits $= 10000$

Sum of the greatest and smallest number of five digits

$$= 99999 + 10000 = 109999$$

2) **Problem 01-01-02:** If x, y, z be the digits of a number beginning from the left, then find the number.

Solution: x, y, z are the digits of a number beginning from the left

$$\Rightarrow \text{The number} = x \times 100 + y \times 10 + z \times 1$$

$$= 100x + 10y + z$$

MCQ

1) **MCQ 01-01-01:** What is the place value of 7 in 92547108?

- (a) 7 (b) 70
(c) 700 (d) 7000

2) **MCQ 01-01-02:** The face value of 6 in the number 5458629 is

- (a) 6 (b) 60
(c) 600 (d) 6000

3) **MCQ 01-01-03:** Every rational number is also

- (a) an integer (b) a real number

(c) a natural number (d) a whole number

4) **MCQ 01-01-04:** The sum of the place values of 5 in the number 503535 is

- (a) 505 (b) 500005
(c) 500505 (d) 50505

5) **MCQ 01-01-05:** If p and q are both odd numbers, which of the following is an even number?

- (a) $p + q$ (b) $p + q + 1$
(c) $pq + 2$ (d) pq

6) **MCQ 01-01-06:** The least prime number is

- (a) 1 (b) 2
(c) 3 (d) 5

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

1) **SAQ 01-01-01: Calculate:** $1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + \dots + 100^2$

Solution: $1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + \dots + 100^2$

= Sum of squares of first 100 natural numbers

$$\begin{aligned} &= \frac{n(n+1)(2n+1)}{6} \text{ where } n = 100 \\ &= \frac{100(100+1)(2 \times 100 + 1)}{6} = \frac{100 \times 101 \times 201}{6} = \mathbf{3,38,350.} \end{aligned}$$

2) **SAQ 01-01-02:** $1 + 2 + 3 + 4 + \dots + 1000$

Solution: $1 + 2 + 3 + 4 + \dots + 1000$

= Sum of first 1000 natural numbers

$$\begin{aligned} &= \frac{n(n+1)}{2} \text{ where } n = 1000 \\ &= \frac{1000(1000+1)}{2} \\ &= \frac{1000(1001)}{2} = \mathbf{5,00,500.} \end{aligned}$$

01-01-02: TESTS OF DIVISIBILITY

In mathematics, divisibility is an important concept that helps us determine whether one number can be divided exactly by another without leaving any remainder. The tests of divisibility are simple and quick methods or rules that allow us to check this property without performing actual division.

These rules form the foundation for various arithmetic operations such as simplifying fractions, finding factors and multiples, checking prime numbers, and solving number system problems. By applying divisibility tests, we can easily identify whether a number is divisible by 2, 3, 4, 5, 6, 8, 9, 10, 11, and so on.

Understanding divisibility not only improves numerical accuracy but also helps develop mental calculation skills and logical reasoning. It builds a strong base for higher mathematical concepts such as factorization, L.C.M. and H.C.F., and modular arithmetic.

Before learning the divisibility rules, it is essential to understand some basic mathematical terms like dividend, divisor, quotient, remainder, factor, multiple.

Dividend: The number that is being divided.

Illustration: In $25 \div 5$, the **dividend** is 25.

Divisor: The number by which another number is divided.

Illustration: In $25 \div 5$, the **divisor** is 5.

Quotient: The result obtained when one number is divided by another.

Illustration: $25 \div 5 = 5$, so the **quotient** is 5.

Remainder: The amount left over after division if the number is not exactly divisible.

Illustration: $26 \div 5 = 5$ remainder 1.

General Relation

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

Divisible Number: A number is said to be **divisible by another** if the division leaves no remainder.

Illustration: 24 is **divisible by 6** because $24 \div 6 = 4$ with **no remainder**.

Factor: A number that divides another number exactly.

Illustration: The factors of 12 are 1, 2, 3, 4, 6, 12.

Multiple: The product of a number and any integer.

Illustration: Multiples of 4 are 4, 8, 12, 16, 20, ...

Tests of Divisibility

1. Divisibility By 2:

A number is divisible by 2 if its unit digit is any of 0, 2, 4, 6, 8.

Illustration: 38674 is divisible by 2, while 96943 is not divisible by 2.

2. Divisibility By 3:

A number is divisible by 3 if the sum of its digits is divisible by 3.

Illustration: Consider, 95421, the sum of digits = 21, which is divisible by 3.

$\therefore$ 95421 is divisible by 3.

Whereas, for the number 9247654, sum of digits = 37, which is not divisible by 3.

$\therefore$ 9247654 is not divisible by 3.

3. Divisibility By 9:

A number is divisible by 9 only when the sum of its digits is divisible by 9.

Illustration: Consider, 95427, the sum of digits = 27, which is divisible by 9.

$\therefore$ 95427 is divisible by 9.

Whereas, for the number 9247654, sum of digits = 37, which is not divisible by 9.

$\therefore$ 9247654 is not divisible by 9.

4. Divisibility By 4:

A number is divisible by 4 if the number formed by its last two digits is divisible by 4.

Illustration: 9879336 is divisible by 4, since 36 is divisible by 4.

896141 is not divisible by 4, since 41 is not divisible by 4.

5. Divisibility By 8:

A number is divisible by 8 if the number formed by its last three digits is divisible by 8.

Illustration: In 96789344, the number formed by last 3 digits, namely 344 is divisible by 8.

$\therefore$ 96789344 is divisible by 8.

Whereas in 676486, the number formed by last 3 digits, namely 486 is not divisible by 8.

$\therefore$ 676486 is not divisible by 8.

6. Divisibility By 10:

A number is divisible by 10 if its unit digit is 0.

Illustration: 9320 is divisible by 10, since its unit digit is 0.

1248405 is not divisible by 10, since its unit digit is not 0.

7. Divisibility By 5:

A number is divisible by 5 only when its unit digit is 0 or 5.

8. Divisibility By 11:

A number is divisible by 11 if the difference between the sum of its digits at odd places and the sum of its digits at even places is either 0 or a number divisible by 11.

Illustration: Consider the number 49231853.

(Sum of its digits at odd places) – (Sum of its digits at even places)

$= (3 + 8 + 3 + 9) - (5 + 1 + 2 + 4) = (23 - 12) = 11$, which is divisible by 11.

$\therefore$ 49231853 is divisible by 11.

Now, Consider the number 66554731.

(Sum of its digits at odd places) – (Sum of its digits at even places)

$= (1 + 7 + 5 + 6) - (3 + 4 + 5 + 6) = (19 - 18) = 1$, which is not divisible by 11.

$\therefore$ 66554731 is not divisible by 11.

9. Divisibility By 25:

A number is divisible by 25 if the number formed by its last two digits is either 00 or divisible by 25.

Illustration: 98475 is divisible by 25 as the number formed by last 2 digits 75 is divisible by 25.

$\therefore$ 98475 is divisible by 25.

10. Divisibility By 7 or 13:

Divide the number into groups of 3 digits (starting from right) and find the difference between the sum of the numbers in odd and even places. If the difference is 0 or divisible by 7 or 13 (as the case may be), it is divisible by 7 or 13.

Illustration: 6547884 $\rightarrow$ 6 / 547 / 884

$(884 + 6) - 547 = 343$, which is divisible by 7 but not by 13.

$\therefore 6547884$ is divisible by 7 and not by 13.

$$540488 \rightarrow 540 / 488$$

$540 - 488 = 52$, which is divisible by 13 but not by 7.

$\therefore 540488$ is divisible by both 13 and not by 7.

11. Divisibility By 16:

A number is divisible by 16, if the number formed by its last 4 digits is divisible by 16.

Illustration: Consider 7891648, the number formed by last 4 digits is 1648, is divisible by 16.

$\therefore 7891648$ is divisible by 16.

12. Divisibility By 6: A number is divisible by 6, if it is divisible by both 2 and 3.

13. Divisibility By 12: A number is divisible by 12, if it is divisible by both 3 and 4.

14. Divisibility By 15: A number is divisible by 15, if it is divisible by both 3 and 5.

15. Divisibility By 18: A number is divisible by 18, if it is divisible by both 2 and 9.

16. Divisibility By 14: A number is divisible by 14, if it is divisible by both 2 and 7.

17. Divisibility By 24: A given number is divisible by 24, if it is divisible by both 3 and 8.

18. Divisibility By 40: A given number is divisible by 40, if it is divisible by both 5 and 8.

19. Divisibility By 80: A given number is divisible by 80, if it is divisible by both 5 and 16.

Note: A number m is divisible by n , if $n = pq$, where p and q are co-primes and m is divisible by both p and q .

Illustration: 398475 is divisible by 75.

Since, $75 = 3 \times 25$, where 3 and 25 are co-primes and 398475 is divisible by both 3 and 25.

If p and q are not co-primes, then the given number m need not be divisible by $n = pq$, even when it is divisible by both p and q .

Illustration: 12 is divisible by both 2 and 4, but it is not divisible by $(2 \times 4) = 8$, since 2 and 4 are not co-primes

SOLVED PROBLEMS

3) Problem 01-01-03: What least value must be assigned to * so that the number 495*1462 is divisible by 3?

Solution: Let the missing digit be x .

Sum of digits = $(4 + 9 + 5 + x + 1 + 4 + 6 + 2) = (31 + x)$.

For $(31 + x)$ to be divisible by 3, x must be replaced by 2.

Hence, the digit in place of * must be 2.

4) Problem 01-01-04: Show that 8193724 is divisible by 11.

Solution: (Sum of digits at odd places) – (Sum of digits at even places) = $(4+7+9+8) - (2 + 3 + 1) = 22$, which is divisible by 11.

Hence, 8193724 is divisible by 11.

MCQ

7) MCQ 01-01-07: Which one of the following numbers is divisible by 3?

- (a) 4006021 (b) 2345679
(c) 2876420 (d) 9566203

8) MCQ 01-01-08: Which one of the following numbers is divisible by 11

- (a) 5823718 (b) 4832718
(c) 9423718 (d) 8432718

9) MCQ 01-01-09: 7386038 is divisible by

- (a) 3 (b) 4
(c) 9 (d) 11

10) MCQ 01-01-10: Which of the following numbers is a multiple of 8?

- (a) 923872 (b) 923972
(c) 923862 (d) 923962

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

3) **SAQ 01-01-03:** Which of the following numbers is divisible by 4?

i. 9734524

ii. 7548693

Solution: Disability test of 4: A number is divisible by 4 if the number formed by its last two digits is divisible by 4.

i. $9734524 \Rightarrow$ Number formed by its last two digits = 24

24 is divisible by 4 $\Rightarrow$ 9734524 is divisible by 4

ii. $7548693 \Rightarrow$ Number formed by its last two digits = 93

93 is not divisible by 4 $\Rightarrow$ 7548693 is not divisible by 4

4) **SAQ 01-01-04:** Is 52563816 divisible by 24?

Solution: $24 = 3 \times 8$, where 3 and 8 are co-primes.

The sum of the digits in the given number is 36, which is divisible by 3. So, the given number is divisible by 3.

The number formed by the last 3 digits of the given number is 816, which is divisible by 8. So, the given number is divisible by 8.

Thus, the given number is divisible by both 3 and 8, where 3 and 8 are co-primes.

So, it is divisible by 3×8 , i.e. 24.

SUMMARY

This unit introduces the fundamental classification of numbers and essential rules of divisibility, which are crucial for solving problems efficiently in competitive examinations. It begins with the study of **types of numbers**, including natural numbers, whole numbers, integers, rational and irrational numbers, real numbers, prime and composite numbers, even and odd numbers, and co-prime numbers. Understanding these categories helps in identifying number properties and relationships.

The unit further covers **tests of divisibility**, which provide quick methods to determine whether a number is divisible by another without performing long division. Key rules include divisibility criteria for numbers such as 2, 3, 4, 5, 6, 8, 9, 10, and 11. Mastery of these tests aid in simplifying calculations, factorization, finding H.C.F. and L.C.M., and solving arithmetic problems rapidly.

Overall, this unit builds a strong numerical foundation and enhances problem-solving speed and accuracy, both of which are essential for competitive exams.

Divisibility Tests:

Test	Condition
Divisibility By 2	If its unit digit is any of 0, 2, 4, 6, 8
Divisibility By 3	If the sum of its digits is divisible by 3
Divisibility By 9	If the sum of its digits is divisible by 9
Divisibility By 4	If the number formed by its last two digits is divisible by 4
Divisibility By 8	If the number formed by its last three digits is divisible by 8
Divisibility By 10	If its unit digit is 0
Divisibility By 5	If its unit digit is 0 or 5
Divisibility By 11	If the difference between the sum of its digits at odd places and the sum of its digits at even places is either 0 or a number divisible by 11
Divisibility By 25	If the number formed by its last two digits is either 00 or divisible by 25.
Divisibility By 7	Divide the number into groups of 3 digits (starting from right) and find the difference between the sum of the numbers in odd and even places. If the difference is 0 or divisible by 7, it is

	divisible by 7
Divisibility by 13	Divide the number into groups of 3 digits (starting from right) and find the difference between the sum of the numbers in odd and even places. If the difference is 0 or divisible by 13, it is divisible by 13
Divisibility By 16	If the number formed by its last 4 digits is divisible by 16.

Combined divisibility criteria:

Test	If it is divisible by both
Divisibility By 6	2 and 3
Divisibility By 12	3 and 4
Divisibility By 15	3 and 5
Divisibility By 18	2 and 9
Divisibility By 14	2 and 7
Divisibility By 24	3 and 8
Divisibility By 40	5 and 8
Divisibility By 80	5 and 16

END OF UNIT EXERCISES

- Which of the following numbers is divisible by 99?
 - 114345
 - 813464
- Calculate: $1 + 2^3 + 3^3 + \dots + 50^3$
- Find the smallest digit * such that $78*3945$ is divisible by 11.
- If a number $476**0$ is divisible by both 3 and 11, then find the non-zero digits in the hundred's and ten's place.

5) What is the sum of all the prime numbers from 1 to 20?

KEY WORDS

Real Numbers, Natural Numbers, Whole Numbers, Integers, Rational Numbers, Irrational Numbers, Imaginary Numbers, Prime Numbers, Even Numbers, Odd Numbers, Divisibility

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UNIT 01-02: H.C.F. AND L.C.M. OF NUMBERS

LEARNING OBJECTIVES

After successful completion of this unit, you will be able to

- ❖ Define and differentiate between factors and multiples.
- ❖ Understand the concepts of Highest Common Factor (H.C.F.) and Least Common Multiple (L.C.M.).
- ❖ Apply different methods for finding H.C.F. and L.C.M. of numbers.
- ❖ Solve competitive-level problems using shortcut and analytical approaches.

INTRODUCTION

In arithmetic and algebra, the concepts of factors and multiples play a crucial role in simplifying numerical and algebraic expressions. The Highest Common Factor (H.C.F.) and the Least Common Multiple (L.C.M.) are fundamental tools used to compare, simplify, and solve problems involving numbers and expressions. These concepts have significant applications in problems related to ratios, time, work, and in the simplification of fractions and algebraic equations.

01-02-01: HIGHEST COMMON FACTOR (H.C.F.) OR GREATEST COMMON DIVISOR (G.C.D.)

Factors and Multiples: If a number a divides another number b exactly, we say that a is a *factor* of b . In this case, b is called a *multiple* of a .

Illustration: 6 divides 120 exactly.

$\Rightarrow$ 6 is a factor of 120 and 120 is a multiple of 6.

Common Factor and Common Multiple:

A number or expression that divides means is a factor of two or more numbers exactly, is called their Common Factor.

A number or expression that is exactly divisible by two or more given numbers is called their Common Multiple.

Prime Factorisation: Prime Factorisation is the process of expressing a given number as a product of its prime factors.

Illustration: Factors of 12 are 1, 2, 3, 4, 6, 12

Here, $4 = 2^2$, $6 = 3 \times 2$, $12 = 2^2 \times 3$

Therefore, the Distinct Prime Factors of 12 are 2 and 3.

Prime Factorisation of $12 = (2^2) \times 3$.

Highest Common Factor (H.C.F.):

The H.C.F. or G.C.D. of two or more numbers is the greatest number that divides each of them exactly.

Illustration: H.C.F. of 12 & 18:

Factors of 12 are 1, 2, 3, 4, 6, 12

Factors of 18 = 1, 2, 3, 6, 9, 18

Common Factors of 12 and 18 are 1, 2, 3, 6

$\Rightarrow$ **Highest Common Factor (H.C.F) = 6**

Methods to find H.C.F.

1) Factorization Method:

Step 1: Express each one of the given numbers as the product of prime factors.

Step 2: Collect the common prime factors with the least powers common to all the numbers.

Step 3: Take the product of the least powers of common prime factors, which gives the H.C.F.

Illustration: To find the H.C.F. of 12 and 18.

Prime factorisation of 12 and 18:

$$12 = 2^2 \times 3$$

$$18 = 2 \times 3^2$$

Common prime factors of 12 and 18 with the least powers common to both: 2 and 3

H.C.F. = Product of Common Prime Factors with the least powers

$$\text{H.C.F.} = 2 \times 3 = 6$$

2) Division Method:

Step 1: For the given two numbers, divide the larger number by the smaller one.

Step 2: Now, divide the divisor by the remainder.

Step 3: Repeat the process of dividing the preceding number by the remainder last obtained till zero is obtained as the remainder. The last divisor is the required H.C.F.

3) Finding the H.C.F. of more than two numbers:

To find the H.C.F. of three numbers, first, we calculate the H.C.F. of any two of those numbers. Then, we take that result and find the H.C.F. with the third number. This process will give us the H.C.F. of all three numbers.

Similarly, to determine the H.C.F. of more than three numbers, we can continue this method by pairing them and finding the H.C.F. step by step.

Illustration: To find H.C.F. of 1120, 840, 560.

$$1120 > 840 > 560$$

So, first we find the H.C.F. of the two largest numbers (1120 and 840).

$$1120 \div 840 \Rightarrow \text{Quotient} = 1 \text{ and Remainder} = (1120 - 840 \times 1) = 280$$

Now dividing 840 by the remainder 280

$$840 \div 280 \Rightarrow \text{Quotient} = 3 \text{ and Remainder} = (840 - 280 \times 3) = 0$$

Here, the last divisor 280 is the H.C.F. of 1120 and 840.

Further, find the H.C.F. of the result 280 and the next number 560.

$$560 \div 280 \Rightarrow \text{Quotient} = 2 \text{ and Remainder} = (560 - 280 \times 2) = 0$$

Here, the last divisor 280 is the H.C.F. of 280 and 560.

Therefore,

$$\text{H.C.F. of } 560, 840 \text{ and } 1120 = 280$$

Note: Two numbers are said to be co-primes if their H.C.F. is 1.

SOLVED PROBLEMS

1) Problem 01-02-01: Find the H.C.F. of 234, 351, and 585 by the division method.

$$\text{Solution: } 585 > 351 > 234$$

First, we will find the H.C.F. of the two largest numbers (585 and 351).

$$585 \div 351 \Rightarrow \text{Quotient} = 1 \text{ and Remainder} = (585 - 351 \times 1) = 234$$

Now dividing 351 by the remainder 234

$$351 \div 234 \Rightarrow \text{Quotient} = 1 \text{ and Remainder} = (351 - 234 \times 1) = 117$$

Further dividing 234 by the remainder 117

$$234 \div 117 \Rightarrow \text{Quotient} = 2 \text{ and Remainder} = (234 - 117 \times 2) = 0$$

Here, the last divisor 117 is the H.C.F. of 585 and 351.

Finally, we need to find the H.C.F. of the result 117 and the next number 234.

$$234 \div 117 \Rightarrow \text{Quotient} = 2 \text{ and Remainder} = (234 - 117 \times 2) = 0$$

Here, the last divisor 117 is the H.C.F. of 117 and 234.

Therefore, H.C.F. of 585, 351 and 234 = 117

- 2) Problem 01-02-02:** Find the H.C.F. of 108, 288 and 360 by the prime factorization method.

Solution: Prime factorization of 108, 288 and 360:

$$108 = 2^2 \times 3^3$$

$$288 = 2^5 \times 3^2$$

$$360 = 2^3 \times 3^2 \times 5$$

Common prime factors with the common least power of 108, 288 and 306 are 2^2 and 3^2 .

Therefore, H.C.F. of 108, 288 and 360 = $2^2 \times 3^2 = 36$.

MCQ

- 1) MCQ 01-02-01:** Factorisation of 440 is
(a) $2 \times 2 \times 2 \times 5 \times 11$ (b) $5 \times 2 \times 2 \times 13$
(c) $2 \times 3 \times 5 \times 17$ (d) $2 \times 3 \times 5 \times 11$
- 2) MCQ 01-02-02:** The highest common factor of 36 and 84 is
(a) 4 (b) 6
(c) 12 (d) 18
- 3) MCQ 01-02-03:** Which of the following is a pair of co-primes?
(a) (16, 62) (b) (18, 27)
(c) (21, 35) (d) (23, 92)
- 4) MCQ 01-02-04:** The H.C.F. of 3556 and 3444 is
(a) 23 (b) 25
(c) 26 (d) 28

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

1) **SAQ 01-02-01:** Find the H.C.F. of 396, 462, and 858.

Solution: $858 > 462 > 396$

First, we will find the H.C.F. of the two largest numbers (858 and 462).

$$858 \div 462 \Rightarrow \text{Quotient} = 1 \text{ and Remainder} = (858 - 462 \times 1) = 396$$

Now dividing 462 by the remainder 396

$$462 \div 396 \Rightarrow \text{Quotient} = 1 \text{ and Remainder} = (462 - 396 \times 1) = 66$$

Further dividing 396 by the remainder 66

$$396 \div 66 \Rightarrow \text{Quotient} = 6 \text{ and Remainder} = (396 - 66 \times 6) = 0$$

Here, the last divisor 66 is the H.C.F. of 858 and 462.

Finally, we need to find the H.C.F. of the result 66 and the next number 396.

$$396 \div 66 \Rightarrow \text{Quotient} = 6 \text{ and Remainder} = (396 - 66 \times 6) = 0$$

Here, the last divisor 66 is the H.C.F. of 66 and 396.

Therefore, H.C.F. of 858, 462 and 396 is 66.

2) **SAQ 01-02-02:** Find the H.C.F. of 45 and 60 using the Prime Factorization Method.

Solution: Prime Factorization of 45 and 60:

$$45 = 3^2 \times 5$$

$$60 = 2^2 \times 5 \times 3$$

Common Factors are 3, 5.

Therefore, the H.C.F. of 45 and 60 = $3 \times 5 = 15$.

01-02-02: LEAST COMMON MULTIPLE (L.C.M.)

Least Common Multiple (L.C.M.): The least number that is exactly divisible by each one of the given numbers is called their L.C.M.

Illustration: L.C.M. of 12 & 18:

Multiples of 12 = 12, 24, 36, 48, 60, 72, ...

Multiples of 18 = 18, 36, 54, 72, 90, ...

Common Multiples of 12 and 18 = 36, 72, ...

$\Rightarrow$ Least Common Multiple (L.C.M.) = 36

Methods to find L.C.M.

1) Factorization Method of Finding L.C.M.:

Step 1: Factorise each one of the given numbers into a product of prime factors.

Step 2: L.C.M. is the product of the highest powers of all the distinct factors.

Illustration: Let us find the L.C.M. 6, 8, 9 and 12 by using factorization method.

Prime Factorisation of 6, 8, 9 and 12:

$$6 = 2 \times 3$$

$$8 = 2^3$$

$$9 = 3^2$$

$$12 = 2^2 \times 3$$

2 and 3 are the distinct prime factors of 6, 8, 9 and 12 and 2^3 and 3^2 are factors with the highest powers.

Therefore, L.C.M. 6, 8, 9 and 12 = $2^3 \times 3^2 = 72$.

2) Common Division Method of Finding L.C.M.:

Step 1: Arrange the given numbers in a row in either ascending or descending order.

Step 2: Divide by a number that divides exactly at least two of the given numbers, and carry forward the numbers that are not divisible.

Step 3: Repeat the above process until no two numbers are divisible by the same number except 1.

Step 4: Calculate the L.C.M. by taking the product of all the divisors and the undivided numbers.

Illustration: Let us find the L.C.M. of 6, 8, 9 and 12 by using the Common Division Method.

Arranging given numbers in a row in ascending order we get,

	6	8	9	12

Here, we observe that out of the 4 given numbers, 6, 9, 12 are exactly divisible by 3.

So, dividing 6, 9, 12 by 3 and writing the quotient below the corresponding number, keeping 8 as it is:

3	6	8	9	12
	2	8	3	4

Now, consider the numbers obtained after division and repeat the same process
So, divide 2, 8, and 4 by 2, write the quotients below the corresponding numbers, and keep 3 as it is.

3	6	8	9	12
2	2	8	3	4
	1	4	3	2

Now among obtained numbers 1,4,3,2 dividing 4 and 2 by 2,

3	6	8	9	12
2	2	8	3	4
2	1	4	3	2
	1	2	3	1

Now, from the obtained numbers, no two numbers are divisible by the same number other than 1. So, we have to stop the process.

And **L.C.M.** = $3 \times 2 \times 2 \times 2 \times 3 = 72$

Hence, the L.C.M. of 6, 8, 9 and 12 is 72.

Note:

- L.C.M. is the 1st number that occurs in the table of all the numbers given.
- L.C.M. is defined only for positive numbers i.e. either positive integers or positive fractions. L.C.M. for negative numbers or zero is NOT possible.
- Product of two numbers = Product of their H.C.F. and L.C.M.
- Co-primes: Two numbers are said to be co-primes if their H.C.F. is 1.

Therefore, L.C.M. of two Co-primes = Product of those two Co-prime numbers.

SOLVED PROBLEMS

3) Problem 01-02-03: Find the L.C.M. of 24, 36 and 40 by using the Common Division Method.

Solution:

2	24	36	40
2	12	18	20
2	6	9	10
3	3	9	5
	1	3	5

L.C.M. of 24, 36 and 40 = $2 \times 2 \times 2 \times 3 \times 3 \times 5 = 360$.

4) Problem 01-02-04: Find the L.C.M. of 8, 9 and 21 by using Factorization Method.

Solution: Prime factorization of 8, 9 and 21:

$$8 = 2^3$$

$$9 = 3^2$$

$$21 = 3 \times 7$$

L. C. M. of 8, 9 and 21

= Product of distinct prime factors with the highest power

$$= 2^3 \times 3^2 \times 7 = 504$$

MCQ

5) MCQ 01-02-05: L.C.M of 20 and 28 is

(a) 56 (b) 70

(c) 140 (d) 35

6) MCQ 01-02-06: If two numbers a & b are co-prime, their L.C.M is

(a) 1 (b) $a \times b$

(c) $a + b$ (d) a/b

7) MCQ 01-02-07: If the product of two numbers is 324 and their H.C.F. is 3, then their L.C.M. will be

(a) 272 (b) 327

(c) 324 (d) 108

8) MCQ 01-02-08: L.C.M of 5, 10 and 25 is

(a) 50 (b) 75

(c) 25 (d) 100

9) MCQ 01-02-09: The sum of two numbers is 45. Their difference is $\frac{1}{9}$ th of their sum. Their H.C.F. is

(a) 100 (b) 5

(c) 20 (d) 25

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

- 3) **SAQ 01-02-03:** The L.C.M. of two numbers is 12 times their H.C.F. The sum of H.C.F. and L.C.M. is 403. If one number is 93, find the other.

Solution: Let, the H.C.F. of two numbers be x .

Therefore, L.C.M. of two numbers = $12x$.

The sum of H.C.F. and L.C.M. is 403.

$$\Rightarrow x + 12x = 403$$

$$\Rightarrow 13x = 403$$

$$\Rightarrow x = 31$$

Therefore, H.C.F. of two numbers = $x = 31$

L.C.M. of two numbers = $12 \times x = 12 \times 31 = 372$

Product of two numbers = Product of their H.C.F. and L.C.M

$$= 31 \times 372 = 11,532$$

One number is 93 and the product of two numbers is 11,532.

Then, the other number is = $11,532 \div 93 = 124$.

- 4) **SAQ 01-02-04:** Product of two co-prime numbers is 117. Find their L.C.M.

Solution: Product of two co-prime numbers is 117

H.C.F. of co-prime numbers = 1

Product of two numbers = Product of their H.C.F. and L.C.M,

$$117 = 1 \times \text{L.C.M.}$$

Therefore, L.C.M. = 117

01-02-03: H.C.F. AND L.C.M. OF FRACTIONS AND DECIMAL FRACTIONS

H.C.F. and L.C.M. of Fractions:

$$H.C.F. = \frac{\text{H.C.F. of Numerators}}{\text{L.C.M. of Denominators}}$$

$$L.C.M. = \frac{\text{L.C.M. of Numerators}}{\text{H.C.F. of Denominators}}$$

H.C.F. and L.C.M. of Decimal Fractions:

To find H.C.F. or L.C.M. of decimal fractions, we remove decimals first by converting them into whole numbers.

Step 1: Count the number of decimal places in each number.

Step 2: Multiply each number by 10, 100, 1000, etc., to make all whole numbers.

Step 3: Find the H.C.F. or L.C.M. of those whole numbers.

Step 4: Divide the result by the same power of 10 used to remove the decimal.

Note: Application of L.C.M. for comparison of fractions:

To compare fractions and find the greatest or smallest:

Step 1: Find the L.C.M. of the denominators of the given fractions.

Step 2: Convert each fraction into an equivalent fraction with this L.C.M. as the new denominator. Multiply both numerator and denominator by the same number so that the denominator becomes equal to the L.C.M.

Step 3: To compare fractions with the same denominator, look only at the numerators. The fraction with the largest numerator is the greatest, and the one with the smallest numerator is the least.

SOLVED PROBLEMS

5) **Problem 01-02-05:** Find the H.C.F. and L.C.M. of $\frac{2}{3}, \frac{4}{5}, \frac{6}{7}$.

Solution: Numerators $\rightarrow 2, 4, 6$

Prime Factorisation:

$$2 = 2$$

$$4 = 2^2$$

$$6 = 2 \times 3$$

$$\text{H.C.F. of } (2, 4, 6) = 2$$

$$\text{L.C.M. of } (2, 4, 6) = 2^2 \times 3 = 12$$

$$\text{Denominators } \rightarrow 3, 5, 7$$

These are distinct prime numbers.

$$\text{H.C.F. of } (3, 5, 7) = 1$$

$$\text{L.C.M. of } (3, 5, 7) = 3 \times 5 \times 7 = 105$$

$$\text{H.C.F. of } \frac{2}{3}, \frac{4}{5}, \frac{6}{7} = \frac{\text{H.C.F. of Numerators}}{\text{L.C.M. of Denominators}} = \frac{2}{105}$$

$$\text{L.C.M. of } \frac{2}{3}, \frac{4}{5}, \frac{6}{7} = \frac{\text{L.C.M. of Numerators}}{\text{H.C.F. of Denominators}} = \frac{12}{1} = 12$$

6) Problem 01-02-06: Find the H.C.F. and L.C.M. of 0.6, 1.5, and 2.1.

Solution: Multiply 0.6, 1.5, and 2.1 each by 10 to obtain whole numbers 6, 15, 21.

Prime Factorisation of the obtained whole numbers 6, 15, 21:

$$6 = 3 \times 2$$

$$15 = 3 \times 5$$

$$21 = 3 \times 7$$

$$\text{H.C.F. of } (6, 15, 21) = 3$$

$$\text{L.C.M. of } (6, 15, 21) = 2 \times 3 \times 5 \times 7 = 210$$

Since, given decimal fractions 0.6, 1.5 and 2.1 are multiplied by 10 to obtain whole numbers 6, 15, 21, dividing obtained H.C.F. and L.C.M. by 10,

$$\text{H.C.F. of } 0.6, 1.5, 2.1 = \frac{3}{10} = 0.3$$

$$\text{L.C.M. of } 0.6, 1.5, 2.1 = \frac{210}{10} = 21.0$$

MCQ

10) MCQ 01-02-10: H.C.F. of $\frac{3}{4}, \frac{9}{8}$

(a) $\frac{3}{4}$

(b) $\frac{9}{8}$

(c) $\frac{9}{4}$

(d) $\frac{3}{8}$

11) MCQ 01-02-11: L.C.M. of $\frac{2}{3}, \frac{4}{9}$

(a) $\frac{4}{9}$

(b) $\frac{4}{3}$

(c) $\frac{2}{9}$

(d) $\frac{2}{3}$

12) MCQ 01-02-12: H.C.F. of 0.3 and 0.9.

(a) 0.9

(b) 0.3

(c) 0.1

(d) 2.7

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

5) SAQ 01-02-05: Find H.C.F. and L.C.M. of $\frac{3}{8}, \frac{9}{20}$.

Solution: Numerators $\rightarrow 3, 9$

Prime Factorisation of 3 and 9:

$$3 = 3$$

$$9 = 3^2$$

$$\text{H.C.F. of } (3, 9) = 3$$

$$\text{L.C.M. of } (3, 9) = 3^2 = 9$$

$$\text{Denominators} \rightarrow 8, 20$$

Prime Factorisation of 8 and 20:

$$8 = 2^3$$

$$20 = 2^2 \times 5$$

$$\text{H.C.F. of } (8, 20) = 2^2 = 4$$

$$\text{L.C.M. of } (8, 20) = 2^3 \times 5 = 40$$

$$\text{H.C.F. of } \frac{3}{8}, \frac{9}{20} = \frac{\text{H.C.F. of Numerators}}{\text{L.C.M. of Denominators}} = \frac{3}{40}$$

$$\text{L.C.M. of } \frac{3}{8}, \frac{9}{20} = \frac{\text{L.C.M. of Numerators}}{\text{H.C.F. of Denominators}} = \frac{9}{4}$$

6) SAQ 01-02-06: Find the H.C.F. of 0.48, 1.6 and 6.4.

Solution: After Multiplying 0.48, 1.6 and 6.4 by 100 obtained whole numbers are 48, 160, 640.

H.C.F. of 0.48, 1.6 and 6.4

$$= \frac{\text{H.C.F. of 48, 160 and 640}}{100}$$

To find H.C.F. of 48, 160 and 640:

$$48 = 2^4 \times 3$$

$$160 = 2^5 \times 5$$

$$640 = 2^7 \times 5$$

$$\text{H.C.F. of 48, 160 and 640} = 2^4 = 16$$

Therefore, H.C.F. of 0.48, 1.6 and 6.4

$$= \frac{\text{H.C.F. of 48, 160 and 640}}{100} = \frac{16}{100} = 0.16$$

SUMMARY

➤ Highest Common Factor (H.C.F.)

- A factor of a number divides it exactly.
- A common factor is a number that divides two or more numbers exactly.
- It is the largest number that divides all given numbers exactly.

➤ Methods of Finding H.C.F.

- Prime Factorization Method
Express each number as prime factors and take the product of Common prime factors with the common least power.
- Division Method (Euclid's Algorithm)
Divide the larger number by the smaller; continue dividing the remainder until zero appears.
Last divisor = H.C.F.
- Least Common Multiple (L.C.M.)
 - A multiple is obtained by multiplying a number.
 - A common multiple is a number that is a multiple of two or more numbers.
 - L.C.M. is the smallest common multiple of two or more numbers.
- Methods of Finding L.C.M.
 - Prime Factorization Method: Product of highest power of all distinct prime factors involved.
 - Common Division Method: Arrange the numbers and divide by a common divisor for at least two until no numbers share a divisor except 1. Then, calculate the L.C.M. by multiplying all divisors and undivided numbers.
- H.C.F. and L.C.M. of Decimal Fractions: Multiply by powers of ten to remove decimals, find the H.C.F. and L.C.M. of whole numbers, and adjust the final answer accordingly.
- Important results:
 - $\text{H.C.F.} \times \text{L.C.M.} = \text{Product of numbers (for two numbers only)}$.
 - If $\text{H.C.F.} = 1 \rightarrow$ numbers are co-prime.

END OF UNIT EXERCISES

- 1) Find the H.C.F. of 884, 1210, and 1596.
- 2) Find the H.C.F. of 72 and 108 using the Prime Factorization Method.
- 3) Find the H.C.F. of 98 and 63 using the Division Method.
- 4) The product of two whole numbers is 1500 and their H.C.F. is 10. Find the L.C.M.
- 5) Find L.C.M. of 14, 28, and 35 using the Prime Factorization Method.

- 6) Find L.C.M. of 9, 12, and 15 using Common Division Method.
- 7) Calculate the H.C.F. of 1.75, 5.6 and 7.
- 8) Find the L.C.M. of $\frac{7}{17}, \frac{5}{53}, \frac{2}{23}, \frac{3}{31}$.

KEY WORDS

Division, Quotient, Divisor, Multiple, Least Common Multiple, Highest Common Factor, Prime Factorization

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UNIT 01-03: SQUARE ROOT AND CUBE ROOT

LEARNING OBJECTIVES

After successful completion of this unit, you will be able to

- ❖ Understand the fundamental concepts of square roots and cube roots.
- ❖ Apply different methods to find square roots and cube roots quickly.
- ❖ Identify perfect squares and perfect cubes using properties and divisibility rules.
- ❖ Solve problems involving square root and cube root in competitive exam contexts.
- ❖ Use short-cut tricks for faster calculation in competitive exams.

INTRODUCTION

In this chapter, we will explore the concepts of square roots and cube roots, essential mathematical operations that play a significant role in various applications, particularly in competitive examinations where efficient problem-solving is vital. Have you ever wondered how to quickly ascertain whether 144 is a perfect square or how to determine the cube root of 12,167 in just seconds without the need for factorization? Mastering these skills not only enhances your mathematical proficiency but also contributes to effective time management during exams. This chapter will provide clear explanations, straightforward methods, effective shortcuts, and plenty of practice opportunities to help you gain confidence in working with square roots and cube roots.

01-03-01: SQUARE ROOTS – BASICS AND METHODS

This section aims to examine the fundamental concepts of squares and square roots, which are integral to various mathematical disciplines. We will begin by defining these terms and exploring the properties of perfect squares, which hold significant importance in number theory. Furthermore, we will analyze several methods for calculating square roots, including prime factorization, division, and approximation techniques. This comprehensive overview is designed to enhance your understanding and enable you to effectively address problems related to squares and square roots.

Definition:

Square: Square of a number is calculated by multiplying a number with itself. That is square of a number is the second power of that number.

If x is any number then, its square is $x * x$ and it is denoted as x^2 .

Illustration: To calculate square of 5.

$$\text{Square of } 5 = 5 * 5 = 5^2 = 25$$

Square Root: If $x^2 = y$, we say that the square root of y is x and we write, $\sqrt{y} = x$.

25 is square of 5 $\Rightarrow$ 5 is a square root of 25.

$$\text{Further, Square of } (-5) = (-5) * (-5) = (-5)^2 = 25$$

Therefore, -5 is also a square root of 25.

Note: Every positive number has two square roots negative and positive. In exams, unless specified, the positive root is taken.

Perfect Square: A perfect square is a number that can be expressed as the square of an integer.

In other words, if m is an integer, then $n = m^2$ is a perfect square.

Properties of Perfect Squares:

1. Unit Digit Rule

- A perfect square can only end with the digits 0, 1, 4, 5, 6, or 9
- It cannot end with the digits 2, 3, 7, or 8.

Example: 49 (ends with 9), 225 (ends with 5).

2. Even Number of Zeros

- A perfect square number that ends with zeros always has those zeros in pairs, resulting in an even count.

Example: $100 = 10^2$ (2 zeros), $10,000 = 100^2$ (4 zeros).

3. Square of Even and Odd Numbers

- Square of an even number $\rightarrow$ always even.
- Square of an odd number $\rightarrow$ always odd.

Example: $12^2 = 144$ (even), $15^2 = 225$ (odd).

4. Numbers Between Perfect Squares

- If $n^2 < x < (n + 1)^2$, then $\sqrt{x}$ lies between n and $(n + 1)$.

5. Sum of Odd Numbers Property

- The sum of the first n odd numbers $= n^2$

Example: Sum of the first 5 odd numbers $= 1 + 3 + 5 + 7 + 9 = 25 = 5^2$.

6. Prime Factorization of Perfect Square

- If a number is a perfect square, then the exponent of each prime factor in its factorization is even.

Example: $144 = 2^4 \times 3^2 \rightarrow$ both exponents are even $\rightarrow$ perfect square.

Methods of Finding Square Roots:

1. Prime Factorization Method:

Step 1: Factorize the number into primes.

Step 2: Group factors into pairs.

Step 3: Multiply one factor from each pair to get the square root.

Illustration: To calculate $\sqrt{144}$ using prime factorization method.

Prime factorization of 144 is

$$144 = 2 * 2 * 2 * 2 * 3 * 3$$

Grouping factors into pairs,

$$144 = (2 * 2) * (2 * 2) * (3 * 3)$$

Multiplying one factor from each pair we get the square root,

$$\sqrt{144} = 2 * 2 * 3 = 12.$$

2. Long Division Method (useful for large numbers):

Step 1: Group digits in pairs from right to left.

Step 2: Find the largest square $\leq$ left-most group.

Step 3: Subtract, bring down the next group.

Step 4: Continue till remainder = 0.

Illustration: Find $\sqrt{1521}$ (by long division).

Pair digits of 1521 from right: (15) (21).

- First group = 15.

Find largest square $\leq 15 \rightarrow 3 \times 3 = 9$.

- Write 3 as the first digit of root.
- Subtract $15 - 9 = 6$.

Bring down next pair (21) $\rightarrow$ new dividend = 621.

Double the root so far ($3 \rightarrow 6$).

- Write as 60_.
- Find digit x such that $(60 + x) \times x \leq 621$.
- Try $x = 9 \rightarrow 69 \times 9 = 621$ (fits exactly).

Place 9 in quotient $\rightarrow$ root now = 39.

- Subtract $621 - 621 = 0$ (remainder zero).

Therefore, $\sqrt{1521} = 39$.

3. Approximation Method:

Step 1: Find the two perfect squares between which the number lies.

Step 2: Approximate the value.

Illustration: Find approximate value of $\sqrt{730}$.

$27^2 = 729$ and $28^2 = 784$ therefore, $\sqrt{730}$ lies between 27 & 28 $\rightarrow \sqrt{730} \approx 27$.

4. Square roots of decimals:

Step 1: Make the decimal places even

Square roots work smoothly when the number of decimal digits is even.

- If digits are odd, add one zero.
- If digits are even, proceed directly.

Step 2: Remove the decimal $\rightarrow$ find square root $\rightarrow$ put decimal back

Number of decimal places in the square root = half the decimal places in the original number.

Illustration: Find: $\sqrt{0.0049}$

Decimal digits = 4

Removing decimal $\rightarrow 49$

$$\sqrt{49} = 7$$

Number of decimal places in the square root = half of 4 = 2

$$\sqrt{0.0049} = 0.07.$$

Note:

$$\text{I. } \sqrt{xy} = \sqrt{x} \sqrt{y}$$

$$\text{II. } \sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}}$$

SOLVED PROBLEMS

- 1) **Problem 01-03-01:** Calculate $\sqrt{225}$ by prime factorization method.

Solution: Prime factorization of 225 is

$$225 = 3 \times 3 \times 5 \times 5 = (3 \times 3) \times (5 \times 5)$$

Therefore, $\sqrt{225} = 3 \times 5 = 15$

- 2) **Problem 01-03-02:** Calculate $\sqrt{4489}$ by long division method.

Solution: Group digits: $\overline{44} \overline{89}$

First group **44**

Largest square ≤ 44 is **36** (6^2)

Quotient so far = **6**

Subtract: $44 - 36 = 8$

Bring down $\rightarrow$ **89**

New Dividend $\rightarrow$ **889**

Double the quotient: **6** $\rightarrow$ **12**

To find digit x such that $(120 + x) \times x \leq 889$

$$121 \times 1 = 121$$

$$122 \times 2 = 244$$

$$123 \times 3 = 369$$

$$124 \times 4 = 496$$

$$125 \times 5 = 625$$

$$126 \times 6 = 756$$

$$127 \times 7 = 889 \text{ (exact)}$$

6	44 89	67
	36	
127	08 89	
	08 89	
	0	

So $\sqrt{4489} = 67$.

MCQ

1) MCQ 01-03-01: $(10)^2 - (3)^2 - 10 = (?)^2$

- (a) 81 (b) 9
(c) 8 (d) 3

2) MCQ 01-03-02: $\frac{\sqrt{196}}{7} \times \frac{\sqrt{441}}{7} \times \frac{80}{\sqrt{225}} =$

- (a) 64 (b) 16
(c) 32 (d) 48

3) MCQ 01-03-03: $\frac{\sqrt{24} + \sqrt{216}}{\sqrt{96}} =$

- (a) $2\sqrt{6}$ (b) 2
(c) $6\sqrt{2}$ (d) $\sqrt{6}$

4) MCQ 01-03-04: $\sqrt{41 - \sqrt{21 + \sqrt{19 - \sqrt{9}}}} =$

- (a) 4 (b) 6
(c) 8 (d) 16

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

1) SAQ 01-03-01: Evaluate: $\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}}$

Solution:
$$\begin{aligned} & \sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}} \\ &= \sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + 15}}}} \quad (\because \sqrt{225} = 15) \\ &= \sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{169}}}} \end{aligned}$$

$$\begin{aligned}
&= \sqrt{10 + \sqrt{25 + \sqrt{108 + 13}}} \quad (\because \sqrt{169} = 13) \\
&= \sqrt{10 + \sqrt{25 + \sqrt{121}}} = \sqrt{10 + \sqrt{25 + 11}} \quad (\because \sqrt{121} = 11) \\
&= \sqrt{10 + \sqrt{36}} \\
&= \sqrt{10 + 6} \quad (\because \sqrt{36} = 6) \\
&= \sqrt{16} = 4
\end{aligned}$$

2) **SAQ 01-03-02:** Find the value of $\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}$.

Solution: Let $\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}} = x$.

$$\text{Then, } \sqrt{2 + x} = x$$

$$\Rightarrow 2 + x = x^2$$

$$\Rightarrow x^2 - x - 2 = 0$$

$$(x - 2)(x + 1) = 0$$

$$x = 2, \quad x = -1$$

$$\Rightarrow x = 2 \quad \because x \neq -1.$$

3) **SAQ 01-03-03:** Find the value of $\sqrt{\frac{0.016}{0.4}}$.

$$\text{Solution: } \sqrt{\frac{0.016}{0.4}} = \sqrt{\frac{0.0016}{0.04}}$$

$$\sqrt{16} = 4 \Rightarrow \sqrt{0.0016} = 0.04$$

$$\sqrt{4} = 2 \Rightarrow \sqrt{0.04} = 0.2$$

$$\therefore \frac{\sqrt{0.0016}}{\sqrt{0.04}} = \frac{0.04}{0.2} = 0.2$$

$$\sqrt{\frac{0.016}{0.4}} = 0.2$$

01-03-02: CUBE ROOTS – BASICS AND METHODS

This section examines cube roots, focusing on their definition, properties, and significance in mathematics. We will explore perfect cubes and the rules governing cube roots, alongside methods for calculation, including prime factorization and the last digit trick for estimation. By understanding these concepts, we aim to enhance our grasp of cube roots and their applications in arithmetic.

Definition:

Cube: Cube of a number is obtained by multiplying a number with itself three times. That is the third power of a number is called its cube.

If x is any number then, its cube is $x * x * x$ and it is denoted as x^3 .

Illustration: To calculate cube of 5.

$$\text{Cube of } 5 = 5 * 5 * 5 = 5^3 = 125$$

Cube Root: If $x^3 = y$, we say that the cube root of y is x and we write, $\sqrt[3]{y} = x$.

125 is cube of 5 $\Rightarrow$ 5 is a cube root of 125.

Methods of Finding Cube Roots:

1. Prime Factorization Method:

Step 1: Factorize the number into primes.

Step 2: Group the factors into triplets means form groups of identical factors—each group must have 3 identical prime factors.

Step 3: Multiply one factor from each triplet to get the cube root.

Illustration: Find $\sqrt[3]{64}$

Prime factorization:

$$64 = 2 \times 2 \times 2 \times 2 \times 2 \times 2$$

$$64 = 2^6$$

Grouping into triplets:

$$(2 \times 2 \times 2) \times (2 \times 2 \times 2)$$

Taking one factor from each triplet and Multiplying:

$$2 \times 2 = 4$$

$$\sqrt[3]{64} = 4$$

2. Shortcut method for large exact cubes

Step 1: The last (unit) digit of the cube — this gives the unit digit of the cube root.

Example: If the cube ends in 8, cube root ends in 2.

Note: Every perfect cube ends with one of the following digits:

Unit digit of cube root	Unit digit of cube
0	0
1	1
2	8
3	7
4	4
5	5
6	6
7	3
8	2
9	9

Step 2: Remove the last three digits. Consider, the remaining number (called leading part).

Step 3: Find the smaller perfect cube nearest to the leading part.

The cube root of that number gives the tens digit of the answer.

Illustration: Find $\sqrt[3]{17576}$.

Unit digit:

Last digit = 6 $\rightarrow$ cube root ends with 6

Removing last 3 digits:

17576 $\rightarrow$ 17

Nearest cube ≤ 17 :

$$2^3 = 8$$

$$3^3 = 27 \text{ (too big)}$$

So, tens digit = 2

Final cube root = 26.

3. Cube roots of decimals:

Step 1: Make decimal digits a multiple of 3

Cube roots work cleanly when decimal digits = 3, 6, 9, ...

If not, add zeros.

Step 2: Remove decimal → find cube root → put decimal back

Number of decimal places in cube root = $\frac{1}{3}$ of decimal places in original number.

Illustration: Find $\sqrt[3]{0.008}$

Decimal digits in 0.008 = 3

Removing decimal → 8

$$\sqrt[3]{8} = 2$$

Number of decimal places in cube root = $3/3 = 1$

$$\sqrt[3]{0.008} = 0.2$$

SOLVED PROBLEMS

3) **Problem 01-03-03:** Calculate $\sqrt[3]{68921}$

Solution: Unit Digit in 68921 is 1 → cube root ends with 1.

Removing last 3 digits:

$$68\textcolor{red}{9}21 \rightarrow 68$$

Nearest cube ≤ 68 :

$$4^3 = 64$$

$$5^3 = 125 \text{ (too big)}$$

So tens digit = 4

$$\sqrt[3]{68921} = 41.$$

4) **Problem 01-03-04:** Find $\sqrt[3]{8000}$

Solution: Prime Factorization of 8000:

$$8000 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 5 \times 5 \times 5$$

$$= (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (5 \times 5 \times 5)$$

$$\sqrt[3]{8000} = 2 \times 2 \times 5 = 20$$

MCQ

5) MCQ 01-03-05: $\sqrt[3]{148877} = ?$

- (a) 49 (b) 59
(c) 63 (d) 53

6) MCQ 01-03-06: The unit digit of 28^3 is

- (a) 4 (b) 2
(c) 6 (d) 9

7) MCQ 01-03-07: If $7^3 = 343$, then the value of $\sqrt[3]{343000}$ is

- (a) 73 (b) 77
(c) 70 (d) 75

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

4) SAQ 01-03-04: Find the smallest number by which 1323 must be multiplied to make it a perfect cube.

Solution: Prime Factorization of $1323 = 3 \times 3 \times 3 \times 7 \times 7$

Here triplet of 7 is missing. So, if we multiply 1323 by 7 will get the triplet of 3 and 7 that is perfect cube of 21.

To make 1323 a perfect cube, it must be multiplied by smallest number 7.

5) SAQ 01-03-05: Evaluate: $\sqrt[3]{64 \times 125 \times 216}$

Solution: $64 \times 125 \times 216 = 4^3 \times 5^3 \times 6^3$

$$\sqrt[3]{64 \times 125 \times 216} = 4 \times 5 \times 6 = 120$$

SUMMARY

This unit covers the concepts of **squares and square roots**, beginning with the meaning of square numbers and progressing to methods of finding square roots through **prime factorization** and the **long division method**, including applications to **decimals** and **fractions**. It further introduces the idea of **cubes and cube roots**, explaining perfect cubes and their properties. The unit details two major methods to find cube roots: the **factorization method**, which involves forming triplets of identical prime factors, and the **shortcut method for exact**

cubes, which uses unit-digit patterns and nearest cube logic. Overall, the unit develops a strong understanding of identifying perfect squares and cubes and calculating their roots accurately using systematic techniques.

END OF UNIT EXERCISES

- 1) Evaluate: $\sqrt{248 + \sqrt{52 + \sqrt{144}}}$.
- 2) Find $\sqrt[3]{79507}$.
- 3) Evaluate: $\sqrt{0.000441}$.
- 4) Evaluate: $\sqrt{0.01 + \sqrt{0.0064}}$.
- 5) Evaluate: $\sqrt{0.01} + \sqrt{0.81} + \sqrt{1.21} + \sqrt{0.16}$.
- 6) Find $\sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}}$.

KEY WORDS

Square, Cube, Square Root, Cube Root, Prime Factorization, Perfect Square, Long Division

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UNIT 01-04: SIMPLIFICATION

LEARNING OBJECTIVES

After successful completion of this unit, you will be able to

- ❖ Apply BODMAS rule accurately to simplify numerical and algebraic expressions.
- ❖ Estimate and approximate results efficiently using rounding and mental math strategies.
- ❖ Use algebraic identities and simplification tricks to solve competitive-level problems.
- ❖ Perform operations on fractions and apply them in problem solving.

INTRODUCTION

In this unit, we will look at important topics that are basic for understanding mathematics, especially for those preparing for competitive exams. We will start with the Order of Operations, known as BODMAS. These rules show the correct order to solve arithmetic problems. They cover Brackets, Orders, Division, Multiplication, Addition, and Subtraction. Next, we will focus on Estimation and Approximation. You will learn techniques like rounding numbers and estimating results to make calculations easier. Finally, we will discuss Basic Algebraic Identities. These identities help simplify expressions and solve equations. Understanding these topics will improve your math skills and problem-solving abilities.

01-04-01: ORDER OF OPERATIONS (BODMAS)

BODMAS stands for Brackets, Orders (Powers/Roots), Division, Multiplication, Addition, and Subtraction. This rule defines the precise sequence for performing operations to obtain correct results.

Order of execution:

1. Brackets – Simplify expressions inside $() \rightarrow \{\} \rightarrow []$ in sequence.
2. Orders – Evaluate powers, roots, or indices.
3. Division and Multiplication – Left to right.
4. Addition and Subtraction – Left to right.

SOLVED PROBLEMS

1) **Problem 01-04-01:** $2048 \div 32 \div 8 = ?$

Solution: In the given expression, we need to perform division twice. We will carry out the operations from left to right.

$$2048 \div 32 \div 8 = \frac{2048}{32} \div 8 = 64 \div 8 = 8$$

2) **Problem 01-04-02:** Simplify: $(425 \times 4000) \div 17000 \times 12$

Solution: To simplify given expression we have to use BODMAS rule. So, we will simplify the multiplication in the bracket first and then division and multiplication from left to right.

Therefore, $(425 \times 4000) \div 17000 \times 12 = 1700000 \div 17000 \times 12 = 100 \times 12 = 1200$.

3) **Problem 01-04-03:** Simplify: $[(84)^2 \div 28 \times 12] \div 24 = 7 \times ?$

Solution: To simplify the left-hand side (LHS) of the given expression using the BODMAS rule, we first need to simplify the expression inside the bracket that is $[(84)^2 \div 28 \times 12]$. To do this, we will follow the order of operations: first, we calculate power, followed by division and multiplication.

Therefore, LHS of expression = $[(84)^2 \div 28 \times 12] \div 24$

$$= [7056 \div 28 \times 12] \div 24$$

$$= [252 \times 12] \div 24 = 126$$

Let, RHS of expression = $7 \times ? = 7 \times x$

Therefore, $126 = 7 \times x$

$$\Rightarrow x = \frac{126}{7}$$

$$\Rightarrow x = 18$$

MCQ

1) **MCQ 01-04-01:** If $45 - [28 - \{37 - (15 - x)\}] = 58$, then x is equal to:

(a) -29 (b) -19

(c) 19 (d) 29

2) MCQ 01-04-02: $4848 \div 24 \times 11 - 222 = ?$

- (a) 200 (b) 2000
(c) 2444 (d) $115\frac{3}{8}$

3) MCQ 01-04-03: $999 \times 99 \times 9 \div 99 \div 9 \div 3 = ?$

- (a) 99 (b) 111
(c) 333 (d) 999

4) MCQ 01-04-04: $((64 - 38) \times 4) \div 13 = ?$

- (a) 4 (b) 1
(c) 8 (d) 2

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

1) SAQ 01-04-01: $(? - 2763) \div 150 \times 12 = 192$

Solution: Let, $(x - 2763) \div 150 \times 12 = 192$

$$\Rightarrow \frac{(x - 2763)}{150} \times 12 = 192$$

$$\Rightarrow \frac{(x - 2763)}{150} = \frac{192}{12}$$

$$\Rightarrow \frac{(x - 2763)}{150} = 16$$

$$\Rightarrow (x - 2763) = 16 \times 150$$

$$\Rightarrow (x - 2763) = 2400$$

$$\Rightarrow x = 2400 + 2763 = 5163.$$

2) SAQ 01-04-02: Simplify: $1 \div [1 + 1 \div \{1 + 1 \div 1(1 + 1 \div 3)\}] + 1$

Solution: $1 \div [1 + 1 \div \{1 + 1 \div 1(1 + 1 \div 3)\}] + 1$

$$= 1 \div \left[1 + 1 \div \left\{ 1 + 1 \div 1 \left(1 + \frac{1}{3} \right) \right\} \right] + 1$$

$$= 1 \div \left[1 + 1 \div \left\{ 1 + 1 \div \left(\frac{4}{3} \right) \right\} \right] + 1$$

$$= 1 \div \left[1 + 1 \div \left\{ 1 + 1 \times \left(\frac{3}{4} \right) \right\} \right] + 1$$

$$\begin{aligned}
&= 1 \div \left[1 + 1 \div \left\{ 1 + \frac{3}{4} \right\} \right] + 1 \\
&= 1 \div \left[1 + 1 \div \frac{7}{4} \right] + 1 \\
&= 1 \div \left[1 + 1 \times \frac{4}{7} \right] + 1 \\
&= 1 \div \left[1 + \frac{4}{7} \right] + 1 \\
&= 1 \div \left[\frac{11}{7} \right] + 1 \\
&= 1 \times \frac{7}{11} + 1 \\
&= \frac{7}{11} + 1 \\
&= \frac{18}{11}.
\end{aligned}$$

01-04-02: ROUNDING NUMBERS AND ESTIMATING RESULTS

Rounding numbers and estimating results are essential skills used to simplify calculations and obtain quick, approximate answers. In many real-life situations and competitive exams, exact calculations are not always necessary; instead, we rely on rounded values to make mental math faster and easier. Rounding replaces a number with a nearby, simpler value, while estimation uses rounding and compatible numbers to quickly approximate sums, differences, products, or quotients. These techniques help check the correctness of answers, solve MCQs efficiently, and make numerical problem-solving more practical and time-saving.

Rounding Numbers

Rounding means replacing a number with another number that is simpler but close to the original value.

Rounding is a form of approximation.

General Rule for Rounding:

Let the digit to be removed be d :

- If $d \geq 5$, increase the previous digit by 1.
- If $d \leq 4$, keep the previous digit as it is.
- All digits to the right become 0 (for whole numbers).

1. For Rounding to nearest 10, check the units place.
 $43 \rightarrow 40$ (as $3 \leq 4$ keep the previous digit 4 as it is)
 $87 \rightarrow 90$ (as $7 \geq 5$ increase the previous 8 digit by 1)
2. For Rounding to nearest 100, check the tens place.
 $364 \rightarrow 400$ (as $6 \geq 5$ increase the previous 3 digit by 1)
 $721 \rightarrow 700$ (as $2 \leq 4$ keep the previous digit 7 as it is)
3. For Rounding to nearest 1000, check the hundreds place.
 $5,432 \rightarrow 5,000$ (as $4 \leq 4$ keep the previous digit 5 as it is)
 $8,650 \rightarrow 9,000$ (as $6 \geq 5$ increase the previous 8 digit by 1)
4. For Rounding Decimals, Focus on the place after the rounding place.
Round 4.73 to 1 decimal $\rightarrow 4.7$
Round 9.486 to 2 decimals $\rightarrow 9.49$
Round 2.995 to 2 decimals $\rightarrow 3.00$

Estimating Results

Estimation is used to get quick approximate results, verify the correctness of answers, and solve MCQs efficiently without performing detailed calculations.

Estimation Tricks:

1. Use Compatible Numbers: Choose numbers that work well together.

Illustration: Instead of $49 \times 19 \rightarrow$ use $50 \times 20 = 1000$

Instead of $198 \div 18 \rightarrow$ use $200 \div 20 = 10$

2. Round to Easy Numbers: Before calculating, round values.

Illustration: $67 + 134 \approx 70 + 130 = 200$

$196 \times 48 \approx 200 \times 50 = 10,000$

3. Front-End Estimation: Use the leftmost digits.

Illustration: $4,872 + 3,190 \approx 4,000 + 3,000 = 7,000$

4. Clustering: If numbers are close, use average.

Illustration: Estimate sum of $198 + 202 + 199 + 201$

All $\approx 200 \rightarrow$ sum $\approx 200 \times 4 = 800$

5. Estimating Decimals: Round decimals to nearest whole or 1 decimal place:

Illustration: $3.87 \times 9.12 \approx 4 \times 9 = 36$

SOLVED PROBLEMS

4) **Problem 01-04-04:** Round 786 to nearest 100.

Solution: Tens digit in $786 = 8 \rightarrow \geq 5$

$$\therefore 786 \rightarrow 800$$

5) **Problem 01-04-05:** Round 9.486 to 2 decimal places.

Solution: Third decimal in $9.486 = 6 \rightarrow$ round up

$$9.486 \rightarrow 9.49$$

6) **Problem 01-04-06:** Estimate $356 \div 18$.

Solution: Round both:

$$356 \approx 360$$

$$18 \approx 20$$

$$360 \div 20 = 18$$

MCQ

5) **MCQ 01-04-05:** Find the approximate value of: 48.9×21.2 .

(a) 100 (b) 1000

(c) 1500 (d) 1200

6) **MCQ 01-04-06:** Find the approximate value of: $\frac{1464}{49.2}$.

(a) 30 (b) 40

(c) 50 (d) 20

7) **MCQ 01-04-07:** $2999 + 1001 + 502 \approx$

(a) 5000 (b) 4000

(c) 4500 (d) 4600

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

3) **SAQ 01-04-03:** Find the approximate value of $\sqrt{102} + 999 \div 4.9$.

Solution: Round of given numbers as: $102 \rightarrow 100$

$$999 \rightarrow 1000$$

$$4.9 \rightarrow 5$$

$$\sqrt{102} + 999 \div 4.9 \approx \sqrt{100} + 1000 \div 5 = 10 + 1000 \div 5 = 10 + 200 = 210$$

$$\sqrt{102} + 999 \div 4.9 \approx 210.$$

4) **SAQ 01-04-04:** Find the approximate value of $\sqrt{\left(\frac{59.08 + 4.87}{5.98 - 1.83}\right)}$.

Solution: Round of given numbers as: 59.08 $\rightarrow$ 59

$$4.87 \rightarrow 5$$

$$5.98 \rightarrow 6$$

$$1.83 \rightarrow 2$$

$$\sqrt{\left(\frac{59.68 + 4.87}{5.98 - 1.83}\right)} \approx \sqrt{\frac{(59 + 5)}{(6 - 2)}} = \sqrt{\frac{64}{4}} = \sqrt{16} = 4$$

$$\sqrt{\left(\frac{59.68 + 4.87}{5.98 - 1.83}\right)} \approx 4.$$

01-04-03: FRACTIONS AND OPERATIONS ON FRACTIONS

Fractions are numerical expressions used to represent parts of a whole or ratios between quantities.

It is written as Fraction = $\frac{p}{q} = \frac{\text{Numerator}}{\text{Denominator}}$

where:

- **Numerator (p)** represents the part taken.
- **Denominator (q)** represents the total number of equal parts ($q \neq 0$).

Example: $\frac{3}{4} \Rightarrow$ Three parts out of four equal parts.

Types of Fractions

Type	Definition	Example	Explanation
Proper Fraction	Numerator < Denominator	3/5	Less than 1
Improper Fraction	Numerator $\geq$ Denominator	7/4	Greater than or equal to 1
Mixed Fraction	Combination of a whole number & a proper fraction	2½	2 + ½ = 5/2
Like Fractions	Same denominators	3/8, 5/8	Denominator 8 is same
Unlike Fractions	Different denominators	2/3, 3/5	Denominators differ
Equivalent Fractions	Represent the same value	1/2 = 2/4 = 3/6	Simplify or multiply by same number
Unit Fraction	Numerator = 1	1/3, 1/5	One part of a whole

Operations on Fractions:

1. Addition and Subtraction of Fractions:

A. Like Fractions: Add or subtract numerators directly.

Illustration:

$$\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$$
$$\frac{7}{9} - \frac{4}{9} = \frac{3}{9} = \frac{1}{3}$$

B. Unlike Fractions: Make denominators same using Least Common Multiple (L.C.M.).

Illustration: $\frac{2}{3} + \frac{3}{5}$

L.C.M. of denominators 3 and 5 is 15.

Making denominators same as 15,

$$\frac{2}{3} + \frac{3}{5} = \frac{2}{3} \times \frac{5}{5} + \frac{3}{5} \times \frac{3}{3} = \frac{10}{15} + \frac{9}{15} = \frac{10+9}{15} = \frac{19}{15} = 1\frac{4}{15}$$

Illustration: $\frac{5}{6} - \frac{1}{4}$

L.C.M. of denominators 6 and 4 is 12.

Making denominators same as 12,

$$\frac{5}{6} \times \frac{2}{2} - \frac{1}{4} \times \frac{3}{3} = \frac{10}{12} - \frac{3}{12} = \frac{10-3}{12} = \frac{7}{12}$$

2. Multiplication of Fractions: Multiply the numerators and denominators directly.

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

Illustration:

$$\frac{2}{5} \times \frac{3}{4} = \frac{6}{20} = \frac{3}{10}$$

Illustration:

$$1\frac{1}{2} \times \frac{2}{3} = \frac{3}{2} \times \frac{2}{3} = 1$$

3. Division of Fractions: Multiply the first fraction by the reciprocal of the second.

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{a \times d}{b \times c}$$

Illustration:

$$\frac{3}{4} \div \frac{2}{5} = \frac{3}{4} \times \frac{5}{2} = \frac{15}{8} = 1\frac{7}{8}$$

Illustration:

$$\frac{7}{9} \div \frac{7}{3} = \frac{7}{9} \times \frac{3}{7} = \frac{3}{9} = \frac{1}{3}$$

4. Simplification of Fractions: To simplify, divide both numerator and denominator by their Highest Common Factor (H.C.F.).

Illustration:

$$\frac{12}{16} = \frac{12 \div 4}{16 \div 4} = \frac{3}{4}$$

SOLVED PROBLEMS

7) Problem 01-04-07: Calculate: $\frac{7}{12} - \frac{1}{18}$

Solution:

$$\begin{aligned} \frac{7}{12} - \frac{1}{18} &= \frac{7 \times 3}{12 \times 3} - \frac{1 \times 2}{18 \times 2} \\ &= \frac{21}{36} - \frac{2}{36} = \frac{21 - 2}{36} = \frac{19}{36} \end{aligned}$$

8) Problem 01-04-08: $\frac{2}{3} \div \frac{5}{9} = ?$

Solution:

$$\frac{2}{3} \div \frac{5}{9} = \frac{2}{3} \times \frac{9}{5} = \frac{18}{15}$$

MCQ

8) MCQ 01-04-08: If $\frac{x}{5} = \frac{9}{15}$, then $x = ?$

- (a) 30 (b) 27
(c) 3 (d) 18

9) MCQ 01-04-09: $5/8 \div 15/16 =$

- (a) $\frac{1}{2}$ (b) $\frac{2}{3}$
(c) $\frac{3}{2}$ (d) $\frac{75}{128}$

10) MCQ 01-04-10: Which of the following fractions is greater than $\frac{3}{4}$ and less than $\frac{5}{6}$?

- (a) $\frac{1}{2}$ (b) $\frac{2}{3}$
(c) $\frac{4}{5}$ (d) $\frac{9}{10}$

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

5) SAQ 01-04-05: Solve: $1\frac{1}{2} + 2\frac{2}{7} = 3\frac{1}{2} + ?$

Solution: Let, $3\frac{1}{2} + ? = 3\frac{1}{2} + x$

$$\Rightarrow 1\frac{1}{2} + 2\frac{2}{7} = 3\frac{1}{2} + x$$

$$\Rightarrow x = 1\frac{1}{2} + 2\frac{2}{7} - 3\frac{1}{2}$$

$$\Rightarrow x = \frac{3}{2} + \frac{16}{7} - \frac{7}{2} = \frac{16}{7} - \frac{4}{2}$$

$$= \frac{32 - 28}{14} = \frac{4}{14} = \frac{2}{7}$$

6) SAQ 01-04-06: Solve: $2\frac{2}{9} + 4\frac{1}{18} - 1\frac{1}{2}$

Solution:

$$2\frac{2}{9} + 4\frac{1}{18} - 1\frac{1}{2} = \frac{20}{9} + \frac{73}{18} - \frac{3}{2}$$

$$= \frac{40 + 73}{18} - \frac{3}{2}$$

$$= \frac{113}{18} - \frac{27}{18}$$

$$= \frac{86}{18} = \frac{43}{9} = 4\frac{7}{9}.$$

01-04-04: BASIC ALGEBRAIC IDENTITIES

Standard algebraic identities are pre-derived formulas that simplify the expansion and factorization of algebraic expressions. They help in solving problems quickly without performing lengthy multiplication steps. These identities form the foundation of algebra and are widely used in competitive exams for fast calculation and simplification.

Some Important Algebraic Identities:

1. $(a + b)^2 = a^2 + 2ab + b^2$
2. $(a - b)^2 = a^2 - 2ab + b^2$
3. $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
4. $(a + b)^2 - (a - b)^2 = 4ab$
5. $a^2 - b^2 = (a + b)(a - b)$
6. $(x + a)(x + b) = x^2 + x(a + b) + ab$
7. $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
8. $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
9. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
10. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
11. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
12. $a^3 + b^3 + c^3 = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) + 3abc$

SOLVED PROBLEMS

9) Problem 01-04-09: If $x + y = 15$ and $xy = 56$, then what is the value of $x^2 + y^2$?

Solution: $(x + y)^2 = x^2 + 2xy + y^2$

$$\Rightarrow x^2 + y^2 = (x + y)^2 - 2xy$$

$$\Rightarrow x^2 + y^2 = (15)^2 - 2 \times 56 = 225 - 112 = 113$$

10) Problem 01-04-10: If $a = \frac{x}{x+y}$ and $b = \frac{y}{x-y}$ then find $\frac{ab}{a+b}$.

Solution:

$$ab = \frac{x}{x+y} \times \frac{y}{x-y} = \frac{xy}{x^2 - y^2}$$

$$a + b = \frac{x}{x+y} + \frac{y}{x-y} = \frac{x(x-y) + y(x+y)}{(x+y)(x-y)} = \frac{x^2 - xy + xy + y^2}{x^2 - y^2} = \frac{x^2 + y^2}{x^2 - y^2}$$

$$\frac{ab}{a+b} = \frac{xy}{x^2 - y^2} \div \frac{x^2 + y^2}{x^2 - y^2}$$

$$\frac{ab}{a+b} = \frac{xy}{x^2 - y^2} \times \frac{x^2 - y^2}{x^2 + y^2}$$

$$\frac{ab}{a+b} = \frac{xy}{x^2 + y^2}$$

MCQ

11) MCQ 01-04-11: If $(a - b) = 4$ and $ab = 2$, then $(a^2 + b^2) = ?$

- (a) 10 (b) 12
(c) 20 (d) 18

12) MCQ 01-04-12: The value of $\frac{(a+b)^2}{(a^2-b^2)}$ is

- (a) $\frac{(a+b)}{(a-b)}$ (b) $\frac{(a+2b)}{(a-2b)}$
(c) $\frac{(a+b)}{(a-b)^2}$ (d) $\frac{a^2+b^2}{(a-b)}$

13) MCQ 01-04-13: $\frac{x+y}{x-y} \div \frac{(x+y)^2}{(x^2-y^2)}$

- (a) -1 (b) 1
(c) $x + y$ (d) $x - y$

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

7) SAQ 01-04-07: If $a + b = 2c$, then find the value of $\frac{a}{a-c} + \frac{b}{b-c}$.

Solution:

$$\begin{aligned} \frac{a}{a-c} + \frac{b}{b-c} &= \frac{a(b-c) + b(a-c)}{(a-c)(b-c)} \\ &= \frac{ab - ac + ba - bc}{ab - ac - cb + c^2} \\ &= \frac{2ab - c(a+b)}{ab - c(a+b) + c^2} \end{aligned}$$

$$\begin{aligned}
&= \frac{2ab - c(2c)}{ab - c(2c) + c^2} \quad \because (a + b) = 2c \\
&= \frac{2ab - 2c^2}{ab - 2c^2 + c^2} = \frac{2ab - 2c^2}{ab - c^2} = \frac{2(ab - c^2)}{ab - c^2} = 2
\end{aligned}$$

8) **SAQ 01-04-08:** If $a + b + c = 0$, then find the value of $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$.

Solution:

$$\begin{aligned}
\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab} &= \frac{a^3}{abc} + \frac{b^3}{cab} + \frac{c^3}{abc} = \frac{a^3 + b^3 + c^3}{abc} \\
&= \frac{(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) + 3abc}{abc} \\
(\because a^3 + b^3 + c^3 &= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) + 3abc) \\
&= \frac{3abc}{abc} \quad (\because a + b + c = 0) \\
&= 3.
\end{aligned}$$

SUMMARY

This unit covers essential arithmetic foundations required for accuracy and speed in competitive examinations. It begins with the Order of Operations, explaining the **BODMAS rule** to correctly evaluate expressions involving brackets, powers, division, multiplication, addition, and subtraction. The unit further develops number sense through **Estimation and Approximation**, focusing on **rounding numbers** and quickly estimating results to verify answers and solve problems efficiently. A detailed study of **Fractions and their operations**—comparison, simplification, addition, subtraction, multiplication, and division—enhances computational skills. The unit also introduces **Basic Algebraic Identities**, enabling simplification of algebraic expressions and solving problems systematically. Together, these topics strengthen numerical reasoning, improve calculation accuracy, and build a strong foundation for higher mathematical concepts and competitive exam problem-solving.

END OF UNIT EXERCISES

- 1) $3565 \div 23 + 4675 \div ? = 430$
- 2) Simplify: $108 \div \left(36 \text{ of } \frac{1}{6}\right) + \frac{2}{3} \times 13 \frac{2}{4}$
- 3) Simplify:
 - a. $45 \div (5 \times 3) + 4$
 - b. $20 - [4 + 2(3)]$
- 4) Evaluate: $18 - 6 \div \left(\frac{1}{3}\right) \times \left(\frac{1}{2}\right) + 2^3$
- 5) A boy was asked to multiply a certain number by 52. He multiplied it by 25 and got his answer less than the correct one by 945. Find the number to be multiplied.

KEY WORDS

BODMAS, Brackets, Powers, Division, Multiplication, Addition, Subtraction, Estimation and Approximation, Fractions, Algebraic Identities

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UNIT 02-01: RATIOS AND PROPORTIONS

LEARNING OBJECTIVES

After successful completion of this unit, you will be able to

- ❖ Understand the concept and notation of ratio and proportion.
- ❖ Distinguish between direct, inverse, and continued proportions.
- ❖ Apply proportional reasoning in practical situations like partnerships, mixtures, and speed-distance problems.
- ❖ Solve competitive exam-style numerical questions based on ratio and proportion.
- ❖ Strengthen logical reasoning and quantitative aptitude through problem-solving.

INTRODUCTION

Ratios and proportions are fundamental mathematical concepts that allow us to compare quantities in meaningful ways. From everyday situations to complex mathematical applications, understanding these concepts is crucial for effective quantitative reasoning. A ratio expresses the relationship between two or more quantities, showcasing how much of one thing there is compared to another. For instance, if you have a recipe that requires 2 cups of flour for every 3 cups of sugar, the ratio of flour to sugar is 2:3. This relationship not only highlights the proportion of each ingredient but also ensures that the recipe's flavor and texture remain consistent. Proportions involve the equality of two ratios. When we say two quantities are in proportion, we mean that they have the same relative size. For example, if you mix paint in equal ratios, the resulting color will be consistently the same regardless of the total amount. Proportions are essential in real-world applications, such as scaling recipes, comparing prices, and analyzing data sets. The applications of ratios and proportions are vast and varied. In daily life, we use these concepts when converting currencies, adjusting ingredient amounts in cooking, or determining distances and travel times. In professional contexts, ratios help in financial analysis, resource allocation, and statistical reports. Understanding these principles enhances our ability to make informed decisions and solve practical problems. In conclusion, mastering the basics of ratios and proportions is not just about numbers; it's about developing a logical

framework for understanding relationships between quantities. This unit will guide you through the basic definitions, explore proportionality in various contexts, and provide practical problem-solving techniques that will enhance your quantitative reasoning skills. By the end of this unit, you will be equipped to apply these concepts effectively in real-world situations.

02-01-01: BASICS OF RATIO AND PROPORTION

Ratio: The ratio of two quantities a and b of same kind and in the same units, is the fraction $\frac{a}{b}$ and it is written as $a : b$.

In the ratio $a : b$, a is called as the first term or antecedent and b is called as the second term or consequent.

Illustration: Weight of Raj is 30 kg and weight of Dev is 40 kg, then the ratio of their weights is 30 : 40 that is 3:4.

Note:

1. In ratio both quantities must have the same units.
2. Ratio has no unit, since the units cancel out.
3. The order of terms is important so, $a : b \neq b : a$.
4. The multiplication or division of each term of a ratio by the same non-zero number does not affect the ratio.

Illustration:

$$\frac{4}{5} = \frac{4 \times 2}{5 \times 2} = \frac{8}{10}$$

Illustration:

$$\frac{4}{6} = \frac{4 \div 2}{6 \div 2} = \frac{2}{3}$$

Simplest Ratio: A ratio is in its simplest form when the two terms have no common factors other than 1.

To obtain the simplest form of the given ratio divide the both the terms of ratio having same units by their Highest Common Factor (H.C.F.).

Illustration: Obtain simplest form of $\frac{25}{30}$.

This ratio is not in the simplest form as 25 and 30 has common factors 1, 5.

To obtain its simplest form divide each term of ratio $\frac{25}{30}$ by H.C.F. of 25 and 30 that is 5.

$$\text{Simplest form of } \frac{25}{30} = \frac{25 \div 5}{30 \div 5} = \frac{5}{6}.$$

Further simplifying a ratio means reducing it to the lowest terms by dividing both terms of the ratio by their H.C.F. (Highest Common Factor).

Ratios are useful to compare two quantities provided those have same units and ratio is expressed in the simplest form.

Equivalent Ratios: Two ratios are said to be equivalent if they represent the same comparison.

Equivalent ratios can be obtained by multiplying or dividing both terms by the same non-zero number.

$$a:b = (ka):(kb),$$

where, k is any non-zero constant.

Note:

1. When a total quantity is divided in a given ratio a:b

$$\text{First Part} = \frac{a}{a+b} \times \text{Total} \quad \text{and} \quad \text{Second Part} = \frac{b}{a+b} \times \text{Total}$$

Illustration: A and B share a profit of ₹7200 in the ratio 5:4.

$$\text{A's share} = \frac{a}{a+b} \times \text{Total Profit} = \frac{5}{5+4} \times ₹ 7200 = ₹ 4000$$

$$\text{B's share} = \frac{b}{a+b} \times \text{Total Profit} = \frac{4}{5+4} \times ₹ 7200 = ₹ 3200$$

2. Every ratio $a:b$ can be expressed as a fraction or percentage:

$$\text{Fraction Form of } a:b = \frac{a}{b} \text{ and}$$

$$\text{Percentage Form } a:b = \left(\frac{a}{b} \times 100 \right) \%$$

Illustration: For ratio, 2:5

$$\text{Fraction Form} = \frac{2}{5} = 0.4$$

$$\text{Percentage Form} = \frac{2}{5} \times 100 = 40 \%$$

Proportion: The equality of two ratios is called proportion.

If $a:b = c:d \Rightarrow a:b :: c:d$ and it is said that a, b, c, d are in proportion.

Here, a and d are called extremes, while b and c are called mean terms.

Note: Product of means = Product of extremes.

Thus, $a:b :: c:d \Leftrightarrow (b \times c) = (a \times d)$.

$$\text{Illustration: } \frac{4}{5} = \frac{8}{10} \Rightarrow 4:5 = 8:10 \Rightarrow 4:5 :: 8:10 \Rightarrow 4 \times 10 = 5 \times 8$$

Note:

- If $a : b = c : d$, then d is called the Fourth Proportional to a, b, c .
- If $a : b = b : c$, then c is called the Third Proportional to a and b .
- If $a : b = b : c$, then b is called the Mean Proportional between a and c .

$\therefore$ If b is the Mean Proportional between a and $c \Rightarrow b = \sqrt{ac}$.

SOLVED PROBLEMS

1) Problem 02-01-01: If $8a = 9b$ then find the ratio of $\frac{a}{9}$ to $\frac{b}{8}$.

Solution:

$$\begin{aligned}
 \text{Ratio of } \frac{a}{9} \text{ to } \frac{b}{8} &= \frac{\left(\frac{a}{9}\right)}{\left(\frac{b}{8}\right)} \\
 &= \frac{a}{9} \times \frac{8}{b} = \frac{8a}{9b} \\
 &= \frac{9b}{9b} = \frac{8a}{8a} && (\because 8a = 9b) \\
 &= \frac{1}{1} = 1
 \end{aligned}$$

2) Problem 02-01-02: Five mangoes and four apples cost as much as three mangoes and seven apples. What is the ratio of the cost of one mango to the cost of one apple?

Solution: Let the cost of one mango be ₹ x and the cost of one apple be ₹ y .

Therefore, cost of five mangoes and four apples = $5x + 4y$

And, cost of three mangoes and seven apples = $3x + 7y$

Further, it is given that the five mangoes and four apples cost as much as three mangoes and seven apples.

$$\Rightarrow 5x + 4y = 3x + 7y$$

$$\Rightarrow 5x - 3x = 7y - 4y$$

$$\Rightarrow 2x = 3y$$

$$\Rightarrow \frac{x}{y} = \frac{3}{2}$$

Therefore, ratio of the cost of one mango to the cost of one apple is $3 : 2$.

MCQ

- 1) **MCQ 02-01-01:** If $5 : a :: 20 : 28$, then a is equal to
a) 4 (b) 6
(c) 7 (d) 8
- 2) **MCQ 02-01-02:** A, B and C are boxes containing marbles in the ratio $1 : 2 : 3$. Total number of marbles is 60. The above ratio can be changed to $3 : 4 : 5$ by transferring
(a) 2 marbles from A to B and 1 from C to B
(b) 3 marbles from B to C
(c) 4 marbles from C to B
(d) 5 marbles from C to A
- 3) **MCQ 02-01-03:** What will be the simplest form of the ratio 3 hours : 1 day?
(a) $1 : 3$ (b) $1 : 6$
(c) $1 : 8$ (d) $1 : 25$
- 4) **MCQ 02-01-04:** Which of the following represents $ab = 64$?
(a) $8 : a = 8 : b$ (b) $a : 16 = b : 4$
(c) $a : 8 = b : 8$ (d) $32 : a = b : 2$
- 5) **MCQ 02-01-05:** The fourth proportional to 5, 8, 15 is
(a) 18 (b) 21
(c) 24 (d) 32

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

- 1) **SAQ 02-01-01:** An amount of ₹. 83 is divided among A, B and C such that A gets 7 more than B and B gets 8 more than C. What is the ratio of their shares?

Solution: Let, an amount of ₹. 83 is divided among A, B and C such that C gets ₹. x .

It is given that B gets 8 more than C

$$\Rightarrow B \text{ gets } ₹. (x + 8).$$

Further, A gets 7 more than B

$$\Rightarrow A \text{ gets ₹. } [(x + 8) + 7].$$

$$\Rightarrow A \text{ gets ₹. } (x + 15)$$

$$\text{Therefore, } (x + 15) + (x + 8) + x = 83$$

$$\Rightarrow 3x + 23 = 83$$

$$\Rightarrow 3x = 83 - 23$$

$$\Rightarrow 3x = 60$$

$$\Rightarrow x = 20$$

The ratio of shares of A, B and C

$$= (x + 15) : (x + 8) : x$$

$$= (20 + 15) : (20 + 8) : 20$$

$$= 35 : 28 : 20$$

2) **SAQ 02-01-02:** The ages of Jay and Maya are in the ratio of 5:2 respectively.

After 7 years, the ratio of their ages will be 4:3. What is the age of Jay?

Solution: Let the present ages of Jay and Maya be $5x$ and $2x$ years respectively.

Jay's age 7 years after = $(5x + 7)$ years.

Maya's age 7 years after = $(2x + 7)$ years.

It is given that, after 7 years, the ratio of their ages will be 4:3.

$$\therefore \frac{5x + 7}{2x + 7} = \frac{4}{3}$$

$$\Rightarrow (5x + 7) \times 3 = (2x + 7) \times 4$$

$$\Rightarrow 15x + 21 = 8x + 28$$

$$\Rightarrow 15x - 8x = 28 - 21$$

$$\Rightarrow 7x = 7 \Rightarrow x = 1$$

Present age of Jay = $5x = 5$ years .

02-01-02: COMPARISON AND TYPES OF RATIOS AND PROPORTION

To compare two ratios, we convert them into like ratios by making their second terms (denominators) the same.

Method to compare ratios: $a : b$ and $c : d$

- Write both ratios as fraction form: $\frac{a}{b}$ and $\frac{c}{d}$
- Find L.C.M. of b and d .
- Express both ratios as equivalent ratios with the same denominator as L.C.M.
- Compare the numerators.

Illustration: Compare $3 : 5$ and $4 : 7$

Fraction form: $\frac{3}{5}$ and $\frac{4}{7}$

L.C.M. of 5 and 7 = 35

Expressing both ratios with the same denominator as 35,

$$\frac{3}{5} = \frac{3 \times 7}{5 \times 7} = \frac{21}{35}$$

$$\frac{4}{7} = \frac{4 \times 5}{7 \times 5} = \frac{20}{35}$$

Comparing the numerators, $21 > 20$.

Therefore, $\frac{3}{5} > \frac{4}{7}$ that is $3 : 5 > 4 : 7$.

Cross Multiplication Method to compare $a : b$ and $c : d$

Compare $a \times d$ and $b \times c$.

- If $a \times d > b \times c$ then the first ratio $a : b$ is greater.
- If $a \times d < b \times c$ then the second ratio $c : d$ is greater.

Illustration: Compare $5 : 8$ and $7 : 12$.

$$5 \times 12 = 60, \quad 8 \times 7 = 56$$

$$5 \times 12 > 8 \times 7 \Rightarrow 5 : 8 > 7 : 12.$$

Componendo and Dividendo: It is a shortcut rule used to simplify and compare ratios.

Componendo:

$$\text{If } \frac{a}{b} = \frac{c}{d} \text{ then } \frac{a+b}{b} = \frac{c+d}{d}.$$

Dividendo:

$$\text{If } \frac{a}{b} = \frac{c}{d} \text{ then } \frac{a-b}{b} = \frac{c-d}{d}.$$

Componendo and Dividendo Combined:

$$\text{If } \frac{a}{b} = \frac{c}{d} \text{ then } \frac{a+b}{a-b} = \frac{c+d}{c-d}.$$

Types of Ratios:

1. **Compounded Ratio:** Ratio obtained by multiplying two or more ratios.

If $a : b, c : d$ are the given ratios then their compounded ratio is $ac : bd$.

The compounded ratio of the ratios $a : b, c : d, e : f$ is $ace : bdf$.

2. **Duplicate ratio:** Square of a given ratio.

Duplicate ratio of $a : b$ is $a^2 : b^2$.

3. **Sub-duplicate ratio:** Square-root of a ratio.

Sub-duplicate ratio of $a : b$ is $\sqrt{a} : \sqrt{b}$.

4. **Triplicate ratio:** Cube of a given ratio.

Triplicate ratio of $a : b$ is $a^3 : b^3$.

5. **Sub-triplicate Ratio:** Cube-root of a ratio.

Sub-triplicate ratio of $a : b$ is $\sqrt[3]{a} : \sqrt[3]{b}$.

Direct and Inverse Proportion:

Direct Proportion: Two quantities x and y are said to be in Direct Proportion (or Direct Variation) if:

- When x increases, y increases
- When x decreases, y decreases

If x is directly proportional to y then, $x = ky$ or $\frac{x}{y} = k$, for some constant k and we write, $x \propto y$.

Illustration:

If 5 kg of rice costs ₹200, what is the cost of 8 kg?

Here, Cost $\propto$ Quantity (weight).

Therefore, Ratio of cost and quantity must be constant.

5 kg of rice costs ₹200 and suppose x is the cost of 8 kg.

$$\frac{200}{5} = \frac{x}{8}$$
$$\Rightarrow x = \frac{200}{5} \times 8 = 320.$$

The cost of 8 kg rice is Rs. 320.

Inverse Proportion: Two quantities x and y are said to be in **Inverse Proportion** if:

- When x increases, y decreases
- When x decreases, y increases

If x is inversely proportional to y , then $xy = k$, for some constant k and we write $x \propto \frac{1}{y}$.

Illustration:

A car covers a distance in 5 hours at 60 km/h. How long will it take at 75 km/h?

Here, $Time \propto 1/Speed \Rightarrow Time \times Speed = Constant$

Let, car will take x hours at 75 km/h speed to cover the same distance.

Therefore, $5 \times 60 = x \times 75$

$$\Rightarrow x = \frac{5 \times 60}{75} = 4.$$

Car will take 4 hours at 75 km/h speed to cover the distance.

SOLVED PROBLEMS

3) Problem 02-01-03: Find the compound ratio of 2 : 5 and 3 : 7.

Solution: Compound Ratio of 2 : 5 and 3 : 7

$$= \frac{2 \times 3}{5 \times 7} = \frac{6}{35} = 6:35$$

- 4) Problem 02-01-04:** S varies directly as R and T varies inversely as R. At some particular time, R = 20, S = 40 and T = 10. If S is changed to 20, then find the value of T?

Solution: At some particular time, R = 20, S = 40 and T = 10

S varies directly as R

$$\Rightarrow S \propto R \text{ and } \frac{S}{R} = \text{constant}$$

$$\Rightarrow \frac{S}{R} = \frac{40}{20} = 2$$

T varies inversely as R

$$\Rightarrow T \propto \frac{1}{R} \text{ and } TR = \text{constant}$$

$$\Rightarrow TR = 10 \times 20 = 200$$

If S is changed to 20, then

$$\frac{S}{R} = 2 \Rightarrow \frac{20}{R} = 2 \Rightarrow R = 10.$$

And if R is 10, then $TR = 200 \Rightarrow T \times 10 = 200 \Rightarrow T = 20$.

MCQ

- 6) MCQ 02-01-06:** The intensity of illumination on a surface from a source of light varies inversely as the square of the distance of the surface from the source. The effect of moving a piece of paper 3 times as far from the source is to

- (a) multiply the intensity by 3
- (b) divide the intensity by 3
- (c) multiply the intensity by 9
- (d) divide the intensity by 9

- 7) MCQ 02-01-07:** If A varies directly proportional to C and B also varies directly proportional to C, which one of the following is not correct?

- a) $(A + B) \propto C$
- (b) $(A - B) \propto \frac{1}{C}$
- (c) $(\sqrt{AB}) \propto C$
- (d) $\frac{A}{B} = \text{Constant}$

8) **MCQ 02-01-08:** Triplicate ratio of 2: 3 is

- a) 4: 9 (b) 8: 27
(c) $\sqrt{2}:\sqrt{3}$ (d) $\sqrt[3]{2}:\sqrt[3]{3}$

9) **MCQ 02-01-09:** When a particular number is subtracted from each of 7, 9, 11 and 15, the resulting numbers are in proportion. The number to be subtracted is

- a) 1 (b) 2
(c) 5 (d) 3

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

3) **SAQ 02-01-03:** The falling height of an object is proportional to the square of the time. One object falls 64 cm in 2 sec then in 6 sec from how much height the object will fall?

Solution: Falling height h of an object is proportional to the square of the time t .

$$\Rightarrow h \propto t^2$$

$$\frac{h}{t^2} = \text{constant}$$

It is given that one object falls 64 cm in 2 sec.

$$\Rightarrow \frac{h}{t^2} = \frac{64}{4} = 16$$

$$\therefore \frac{h}{t^2} = \text{constant} = 16$$

$$\text{For, } t = 6, \quad h \text{ will be } \frac{h}{6^2} = 16 \Rightarrow h = 16 \times 36 = 576$$

In 6 sec the object will fall from height 576 cm.

4) **SAQ 02-01-04:** The time of oscillation of a pendulum varies as the square root of its length. If a pendulum of length 50 cm oscillates once in a second, find the length of the pendulum oscillating once in 2.5 seconds.

Solution: Time of oscillation of a pendulum $t \propto \sqrt{\text{length of pendulum } l}$

$$t \propto \sqrt{l}$$

$$\frac{t}{\sqrt{l}} = \text{constant}$$

A pendulum of length 50 cm oscillates once in a second

$$\Rightarrow \frac{t}{\sqrt{l}} = \frac{1}{\sqrt{50}}$$

Therefore, for length l of the pendulum oscillating once in 2.5 seconds we have,

$$\frac{2.5}{\sqrt{l}} = \frac{1}{\sqrt{50}}$$

$$\Rightarrow \sqrt{l} = 2.5 \times \sqrt{50}$$

$$\Rightarrow l = 6.25 \times 50 = 312.5 \text{ cm}$$

SUMMARY

Ratio and Proportion help compare quantities and understand their mathematical relationships. They are widely used in arithmetic, algebra, commercial mathematics, and real-life applications such as speed–time problems, work–time, mixtures, finance, and scaling.

A **ratio** compares two quantities of the same kind using division. If a and b are two quantities, ratio = **a : b**.

- Ratios are simplified using H.C.F..
- They have **no units** and express **relative comparison**, not absolute values.

Types of Ratios

- **Simple Ratio:** Direct comparison of two values (e.g., 3 : 4).
- **Compound Ratio:** Product of two or more ratios $\rightarrow$ **ac : bd**.
- **Duplicate Ratio:** Squares of the terms ($a^2 : b^2$).
- **Sub-duplicate Ratio:** Square roots of terms ($\sqrt{a} : \sqrt{b}$).
- **Triplicate/Sub-triplicate:** Cubes or cube roots of terms.

Comparison of Ratios

Ratios can be compared by:

- Converting to like terms, or

- Using **cross multiplication**: $\frac{a}{b} > \frac{c}{d} \Leftrightarrow ad > bc$

Proportion

Two ratios are in **proportion** when they are equal:

$$a : b = c : d$$

- a and d = **extremes**
- b and c = **means**

Types of Proportion

- **Direct Proportion:** One quantity increases with the other.

$$\frac{x}{y} = \text{constant}$$

- **Inverse Proportion:** One increases while the other decreases.

$$xy = \text{constant}$$

Special Proportion Concepts

- **Third Proportional:** If $a : b = b : c \rightarrow c$ is third proportional.
- **Fourth Proportional:** If $a : b = c : d \rightarrow d$ is fourth proportional.

Componendo and Dividendo

$$\frac{a}{b} = \frac{c}{d} \text{ then } \frac{a+b}{a-b} = \frac{c+d}{c-d}$$

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END OF UNIT EXERCISES

- 1) Seema and Meena divide a sum of Rs. 25000 in the ratio of 3 : 2 respectively. If Rs. 5000 is added to each of their shares, what would be the new ratio formed?
- 2) If one star equals four circles and three circles equal four diamonds, then what is the ratio of star to diamond?
- 3) Express 50 paise: Rs. 4 in its simplest form.
- 4) Raj and Suraj bought a bottle of water together in which there was 3 litres of water. Raj drank $\frac{1}{3}$ rd part of it. The rest water was drunk by Suraj. Find the amount of water used by Suraj.
- 5) Smita works in her office for 5 hours and Kajal works in her office for 7 hours 30 minutes. Find the ratio of their working hours.

KEY WORDS

Simple Ratio, Equivalent Ratios, Compounded Ratio, Duplicate Ratio, Triplicate Ratio, Componendo, Dividendo, Direct Proportion, Inverse Proportion.

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UNIT 02-02: LOGARITHMS

LEARNING OBJECTIVES

After successful completion of this unit, you will be able to

- ❖ Understand the basic concept and definition of logarithms and their historical significance.
- ❖ Establish and interpret the relation between logarithmic and exponential forms.
- ❖ Apply and verify the fundamental laws of logarithms in simplifying expressions.
- ❖ Solve numerical problems involving logarithmic equations and expressions.
- ❖ Develop problem-solving strategies using logarithmic properties for competitive examinations.

INTRODUCTION

In mathematics, logarithms serve as a bridge between basic arithmetic and more advanced algebraic concepts, particularly those involving powers and exponents. They are powerful tools for simplifying large numerical calculations and solving complex exponential equations. A logarithm transforms multiplication into addition, division into subtraction, and powers into simple products, thereby making computations more efficient. Introduced by John Napier in 1614, logarithms revolutionized manual calculation methods long before the advent of calculators and computers. Logarithms play a crucial role in various fields such as science, engineering, computer algorithms, and data analysis.

02-02-01: BASICS OF LOGARITHM

Logarithm - If $a^x = b$, where a is a positive real number and $a \neq 1$, then $x = \log_a b$ that is the value of $\log b$ to the base a is x .

Here,

- a is called the base,
- b is the argument,
- x is the logarithm or index.

Illustration: $2^3 = 8 \Rightarrow \log_2 8 = 3$.

Note: For any logarithm to be valid, the base must be positive and not equal to 1 ($a > 0, a \neq 1$), and the argument must also be positive ($b > 0$).

Relation between Logarithmic and Exponential Form

A logarithmic function is the inverse of an exponential function.

If $b > 0$ and $a > 0, a \neq 1$, then

$x = \log_a b$ if and only if $a^x = b$.

A logarithm is the exponent to which a base must be raised to obtain a given number.

Therefore, it is easy to evaluate simple logarithmic expressions using an understanding of exponents.

Illustrations:

- I. $\log_2 16 = 4$ because $16 = 2^4$
- II. $\log_5 \sqrt{5} = \frac{1}{2}$ because $\sqrt{5} = 5^{\frac{1}{2}}$
- III. $\log_7 1 = 0$ because $1 = 7^0$
- IV. $\log_7 7 = 1$ because $7 = 7^1$
- V. $\log_7 \frac{1}{49} = -2$ because $\frac{1}{49} = \frac{1}{7^2} = 7^{-2}$

SOLVED PROBLEMS

1) Problem 02-02-01: Evaluate: $\log_{100}(0.0001)$

Solution: To evaluate $\log_{100}(0.0001)$ means by the definition of logarithm, we need to find an exponent x of base the 100 such that, $100^x = 0.0001$.

$$\therefore 0.0001 = \frac{1}{10000} = \frac{1}{100^2} = 100^{-2}$$

Hence, $\log_{100}(0.0001) = -2$.

2) Problem 02-02-02: Evaluate: $\log_{11} \sqrt[3]{121}$

Solution: To find $\log_{11} \sqrt[3]{121}$ means by the definition of logarithm, we need to find an exponent x of base the 11 such that, $11^x = \sqrt[3]{121}$.

$$\therefore \sqrt[3]{121} = 121^{\frac{1}{3}} = (11^2)^{\frac{1}{3}} \text{ since, } 121 = 11^2$$

$$\sqrt[3]{121} = 11^{\frac{2}{3}}$$

Hence, $\log_{11} \sqrt[3]{121} = \frac{2}{3}$.

MCQ

1) MCQ 02-02-01: $\log_3 27 =$

- a) 9 (b) 3
(c) 2 (d) 4

2) MCQ 02-02-02: $\log_4 \frac{1}{64}$

- a) 3 (b) -3
(c) 2 (d) 4

3) MCQ 02-02-03: $\log_{100}(0.01)$

- a) 1 (b) -1
(c) 2 (d) -2

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

1) SAQ 02-02-01: Evaluate: $\log_{12} \sqrt[5]{144}$

Solution: $\sqrt[5]{144} = 144^{\frac{1}{5}} = (12^2)^{\frac{1}{5}}$ since, $144 = 12^2$

$$\sqrt[5]{144} = 12^{\frac{2}{5}}$$

$$\log_{12} \sqrt[5]{144} = \frac{2}{5}$$

2) SAQ 02-02-02: Evaluate $\log_6 \frac{1}{1296}$

Solution: $\frac{1}{1296} = \frac{1}{1296} = \frac{1}{6^4}$ since, $1296 = 6^4$

$$\frac{1}{1296} = \frac{1}{6^4} = 6^{-4}$$

$$\log_6 \frac{1}{1296} = \log_6 6^{-4} = -4$$

02-02-02: LAWS OF LOGARITHMS

The properties of logarithms constitute an essential part of mathematical analysis, offering systematic techniques for simplifying expressions involving powers, products, and quotients. These properties establish a set of fundamental rules that transform multiplicative and exponential relationships into additive or linear forms, thereby facilitating easier computation and interpretation. Understanding and applying these laws—such as the product, quotient, and

power rules—enable learners to manipulate logarithmic expressions efficiently and to solve complex equations involving exponents and logarithms. Moreover, these properties serve as a bridge between algebraic operations and exponential functions, forming a crucial foundation for higher studies in mathematics, physics, engineering, and various competitive examinations.

Laws of Logarithm:

If $a > 0, a \neq 1$ and $x > 0, y > 0$ then

1. Product Law: $\log_a(xy) = \log_a x + \log_a y$

Illustration: $\log_{10}(10 \times 5) = \log_{10} 10 + \log_{10} 5$

2. Quotient law: $\log_a \left(\frac{x}{y} \right) = \log_a x - \log_a y$

Illustration: $\log_5 \left(\frac{20}{125} \right) = \log_5 20 - \log_5 125$

3. Power Law: $\log_a(x^m) = m\log_a x$

Illustration: $\log_2(2^5) = 5(\log_2 2)$

4. Change of Base Law: For $b > 0, b \neq 1$

$$\log_a x = \frac{\log_b x}{\log_b a} = \frac{\log x}{\log a}$$

Illustration: $\log_3 5 = \frac{\log_{10} 5}{\log_{10} 3} = \frac{\log 5}{\log 3}$

5. Fundamental Values: $\log_a 1 = 0, \log_a a = 1$

Illustration: $\log_{20} 1 = 0, \log_{18} 18 = 1$

6. $\log_a x = \frac{1}{\log_x a}$

Illustration: $\log_{49} 7 = \frac{1}{\log_7 49}$

7. $a^{\log_a x} = x$

Illustration: $10^{\log_{10} 125} = 125$

Note:

- Logarithms to the base 10 are known as Common logarithms and it is denoted by $\log x$.

- Logarithms to the base e are known as Natural logarithms and it is denoted by $\ln x$.
- $\ln x = 2.303 \log x$
- When base is not mentioned, it is taken as 10.
- The logarithm of a number contains two parts: characteristic and mantissa.
- **Characteristic:** It is the integral part of the logarithm of a number.
 - When the number is greater than 1: Characteristic = (number of digits to the left of the decimal point in the given number -1)

Illustration:

a) For Number 453.78 Characteristic is **2**.

b) For Number 67.16 Characteristic is **1**.

- When the number is less than 1: Characteristic = negative of (number of zeros between the decimal point and the first significant digit of the number + 1)

Illustration:

a) For Number 0.458 Characteristic is **$\bar{1}$** .

b) For Number 0.00186 Characteristic is **$\bar{3}$** .

Negation is denoted by bar notation.

- **Mantissa:** It is the decimal part of the logarithm of a number and it is obtained from logarithmic tables.

SOLVED PROBLEMS

3) Problem 02-02-03: Evaluate: $\log_3 5 \times \log_{125} 81$

Solution: $\log_3 5 \times \log_{125} 81$

To simplify this, make the base same by using the Change of Base Law,

$$\log_3 5 \times \log_{125} 81 = \frac{\log 5}{\log 3} \times \frac{\log 81}{\log 125} = \frac{\log 5}{\log 3} \times \frac{\log 3^4}{\log 5^3}$$

By using power law, $\log 3^4 = 4 \log 3$ and $\log 5^3 = 3 \log 5$

Therefore,

$$\log_3 5 \times \log_{125} 81 = \frac{\log 5}{\log 3} \times \frac{\log 3^4}{\log 5^3} = \frac{\log 5}{\log 3} \times \frac{4 \log 3}{3 \log 5} = \frac{4}{3}$$

4) Problem 02-02-04: Evaluate: $\log_{10}(10 \times 10^2 \times 10^3 \times \dots \times 10^9)$

Solution: $\log_{10}(10 \times 10^2 \times 10^3 \times \dots \times 10^9)$

$$= \log_{10} 10 + \log_{10} 10^2 + \log_{10} 10^3 + \log_{10} 10^4 + \dots + \log_{10} 10^9$$

$$(\text{Since, } \log_a(xy) = \log_a x + \log_a y)$$

$$= \log_{10} 10 + 2 \log_{10} 10 + 3 \log_{10} 10 + 4 \log_{10} 10 + \dots + 9 \log_{10} 10$$

$$(\text{Since, } \log_a(x^m) = m(\log_a x))$$

$$= 1 + 2 + 3 + 4 + \dots + 9 = 45$$

$$(\text{Since, } \log_a a = 1 \Rightarrow \log_{10} 10 = 1)$$

5) Problem 02-02-05: Evaluate: If $\log_{10} 2 = 0.3010$, then calculate the value of $\log_{10} 80$.

Solution: $\log_{10} 80 = \log_{10}(10 \times 8)$

$$= \log_{10} 10 + \log_{10} 8 \quad (\text{Since, } \log_a(xy) = \log_a x + \log_a y)$$

$$= 1 + \log_{10} 2^3 \quad (\text{Since, } \log_a a = 1 \Rightarrow \log_{10} 10 = 1)$$

$$= 1 + 3 \log_{10} 2 \quad (\text{Since, } \log_a(x^m) = m(\log_a x) \Rightarrow \log_{10} 2^3 = 3 \log_{10} 2)$$

$$= 1 + 3 \times 0.3010$$

$$= 1.9030$$

6) Problem 02-02-06: Find the value of $\log_{\sqrt{3}} 81$.

$$\text{Solution: } \log_{\sqrt{3}} 81 = \frac{\log 81}{\log \sqrt{3}} \quad \left(\because \log_a x = \frac{\log x}{\log a} \right)$$

$$= \frac{\log 3^4}{\log 3^{\frac{1}{2}}} = \frac{4 \log 3}{\frac{1}{2} \log 3} \quad (\because \log_a(x^m) = m(\log_a x))$$

$$= 4 \times 2 = 8$$

MCQ

4) MCQ 02-02-04: $\frac{\log \sqrt{12}}{\log 12} =$

(a) $\frac{1}{12}$

(b) $\frac{1}{2}$

(c) $\frac{1}{\sqrt{12}}$ (d) 2

5) **MCQ 02-02-05:** The value of $\frac{8 \log_{10} 10000}{4 \log_{10} 100}$ is equal to

(a) 2 (b) $\frac{1}{2}$

(c) 4 (d) $\frac{1}{4}$

6) **MCQ 02-02-06:** $\frac{1}{\log_2 30} + \frac{1}{\log_3 30} + \frac{1}{\log_5 30} =$

(a) 2 (b) 3

(c) 5 (d) 1

7) **MCQ 02-02-07:** If $\log 27 = 1.431$, then the value of $\log 9$ is

(a) 0.954 (b) 0.945

(c) 0.934 (d) 0.940

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

3) **SAQ 02-02-03: Simplify:** $\left[\frac{1}{\log_{xy}(xyz)} + \frac{1}{\log_{yz}(xyz)} + \frac{1}{\log_{zx}(xyz)} \right]$

Solution: $\left[\frac{1}{\log_{xy}(xyz)} + \frac{1}{\log_{yz}(xyz)} + \frac{1}{\log_{zx}(xyz)} \right]$

$$= \log_{xyz}(xy) + \log_{xyz}(yz) + \log_{xyz}(zx) \quad \left(\because \log_a x = \frac{1}{\log_x a} \right)$$

$$= \log_{xyz}(xy \times yz \times zx) \quad (\because \log_a(xy) = \log_a x + \log_a y)$$

$$= \log_{xyz}(xyz)^2$$

$$= 2 \log_{xyz}(xyz) \quad (\because \log_a(x^m) = m(\log_a x))$$

$$= 2 \times 1 = 2 \quad (\because \log_a a = 1)$$

4) **SAQ 02-02-04:** Evaluate: $\log_5 \log_3(243)$

Solution: $\log_5 \log_3(243) = \log_5 \log_3(3)^5$

$$= \log_5(5 \log_3 3) \quad (\because \log_a(x^m) = m(\log_a x))$$

$$= \log_5(5 \times 1) \quad (\because \log_a a = 1)$$

$$= \log_5(5) = 1 \quad (\because \log_a a = 1)$$

SUMMARY

If $a > 0, a \neq 1$ and $x > 0, y > 0$ then

- $a^x = b \Rightarrow x = \log_a b$
- $\log_a(xy) = \log_a x + \log_a y$
- $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$
- $\log_a(x^m) = m(\log_a x)$
- $\log_a x = \frac{\log_b x}{\log_b a} = \frac{\log x}{\log a}$
- $\log_a 1 = 0, \log_a a = 1$
- $\log_a x = \frac{1}{\log_x a}$
- $a^{\log_a x} = x$

END OF UNIT EXERCISES

- 1) Calculate: $\log_5 \frac{(125)(625)}{25}$
- 2) Calculate: $\log \frac{9}{14} - \log \frac{15}{16} + \log \frac{35}{24}$
- 3) Evaluate: $(\log_5 5)(\log_4 9)(\log_3 2)$
- 4) Evaluate: $\log_2 \log_2 \log_3 \log_3 27^3$
- 5) Evaluate: $\log_2 32 + \log_3 27 - \log_5 25$

KEY WORDS

Logarithm, Base, Argument, Characteristic, Mantissa, Product, Quotient, Power

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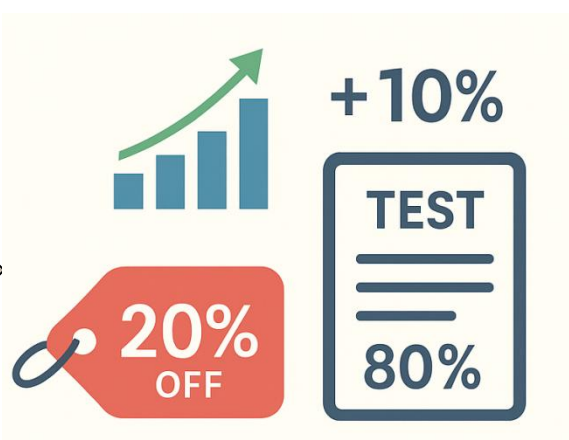
UNIT 02-03: PERCENTAGES

LEARNING OBJECTIVES

After successful completion of this unit, you will be able to

- ❖ Understand the meaning and concept of percentage.
- ❖ Apply basic percentage formulas in numerical problems.
- ❖ Compute percentage increase and decrease effectively.
- ❖ Use shortcut methods for fast calculations in competitive exams.
- ❖ Solve practical problems based on profit, loss, marks, population, and data interpretation using percentages.

INTRODUCTION



Have you ever noticed how discounts, profit margins, or examination results are expressed as percentages? Whether calculating a sale discount, marks obtained, or growth rate, percentages help compare quantities on a common scale of 100.

In competitive exams, percentage problems are frequently asked — often linked with profit & loss, ratio, and data interpretation. Mastering percentage formulas and quick calculation tricks not only improves accuracy but also saves valuable time during exams.

02-03-01: CONCEPT OF PERCENT

We know that a fraction $\frac{1}{3}$ means 1 out of 3 equal parts. $\frac{6}{13}$ means 6 out of 13 equal parts and $\frac{25}{100}$ means 25 out of 100 equal parts.

A fraction whose denominator is 100 or a ratio whose second term is 100 is called as percent.

Illustration: $\frac{75}{100}$ means 75 percent and it is also written as 75%.

To express fraction or decimal as %:

To express any fraction or decimal as % we need to multiply that fraction or decimal with 100 or convert the fraction into equivalent fraction having denominator 100.

Illustration: Fraction $\frac{1}{4}$ as % = $\left(\frac{1}{4} \times 100\right)\% = 25\%$ **OR**

$$\frac{1}{4} = \left(\frac{1}{4} \times \frac{25}{25}\right) = \frac{25}{100} = 25\%$$

$$0.75 \text{ as \%} = 0.75 \times 100 = 75$$

To express % as fraction or decimal:

To express % as fraction or decimal we need to divide it by 100.

$$30 \% = \frac{30}{100} = \frac{3}{10}$$

$$3.5 \% = \frac{3.5}{100} = 0.035$$

Note:

$$1. \ x \% \text{ of } y = \frac{x}{100} \times y$$

$$\textbf{Illustration: } 30 \% \text{ of } 300 = \frac{30}{100} \times 300 = 90$$

$$2. \ x \% \text{ of } y = y \% \text{ of } x$$

$$3. \text{ Percentage} = \frac{\text{Value}}{\text{Total Value}} \times 100$$

Illustration: Scoring 40 marks out of 50 means:

$$\frac{40}{50} \times 100 = 80\%$$

SOLVED PROBLEMS

1) Problem 02-03-01: Express each of the following as a fraction.

- i. 76% ii. 10% iii. 0.8% iv. 0.06%

Solution:

- i. $76\% = \frac{76}{100} = \frac{19}{25}$
ii. $10\% = \frac{10}{100} = \frac{1}{10}$
iii. $0.8\% = \frac{0.8}{100} = \frac{8}{1000} = \frac{1}{125}$
iv. $0.06\% = \frac{0.06}{100} = \frac{6}{10000} = \frac{3}{5000}$

2) Problem 02-03-02: Express each of the following as a decimal.

- i. 9% ii. 23% iii. 0.3% iv. $\frac{1}{2}\%$

Solution:

- i. $9\% = \frac{9}{100} = 0.09$
ii. $23\% = \frac{23}{100} = 0.23$
iii. $0.3\% = \frac{0.3}{100} = 0.003$
iv. $\frac{1}{2}\% = \frac{1}{2} \times \frac{1}{100} = \frac{1}{200} = 0.005$

MCQ

1) MCQ 02-03-01: How is $\frac{2}{5}$ expressed as percentage?

- (a) 0.4 % (b) 4 %
(c) 0.04 % (d) 40 %

2) MCQ 02-03-02: 12% of 6000 =?

- (a) 72 (b) 720
(c) 120 (d) 7200

3) MCQ 02-03-03: 0.01 is what percent of 0.1?

- (a) $\frac{1}{10}\%$ (b) 10 %
(c) $\frac{1}{100}\%$ (d) 100 %

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

- 1) **SAQ 02-03-01:** 30% mangoes out of 550 are rotten. How many mangoes are in good condition?

Solution: 30% mangoes out of 550 are rotten

$$\Rightarrow \text{Number of rotten mangoes} = 30 \% \text{ of } 550 = \frac{30}{100} \times 550 = 165 =$$

$\therefore$ Number of mangoes in good condition

= Total number of Mangoes – Number of rotten mangoes

$$= 550 - 165 = 385$$

- 2) **SAQ 02-03-02:** 4 kg of metal A and 16 kg of metal B are mixed to form an alloy. Then calculate the percentage of metal A in the alloy.

Solution: Weight of metal A in alloy = 4 kg

Weight of metal B in alloy = 16 kg

Total weight of alloy = (4 + 16) kg = 20 kg

$$\text{Percentage of metal A in the alloy} = \frac{\text{Weight of metal A in alloy}}{\text{Total weight of alloy}} \times 100$$

$$= \left(\frac{4}{20} \times 100\right) \% = \left(\frac{1}{5} \times 100\right) \% = 20\%$$

02-03-02: PERCENTAGE CHANGE, INCREASE AND DECREASE

In our daily life, we often compare new values with old ones — such as the rise in petrol prices, an increase in our test scores, or a discount on a product. These comparisons are expressed as percentage change.

For example, if a packet of milk that cost ₹50 yesterday costs ₹55 today, it means the price has increased by 10%. Similarly, if a shirt's price falls from ₹800 to ₹640 during a sale, it represents a 20% decrease.

Understanding percentage change helps us quickly measure how much something has grown or reduced compared to its original value — a concept used widely in economics, business, and everyday decision-making.

Percentage Change: Percentage change shows how much a quantity increases or decreases compared to its original value.

$$\text{Percentage Change} = \frac{\text{Change}}{\text{Original Value}} \times 100$$

If a quantity increase compared to its original value, then the

$$\text{Percentage Increase} = \frac{\text{Increase}}{\text{Original Value}} \times 100$$

If a quantity decrease compared to its original value, then the

$$\text{Percentage Decrease} = \frac{\text{Decrease}}{\text{Original Value}} \times 100$$

Note:

If a certain value V increases by $x\%$, then increased new value $= (100 + x)\%$ of V

If a certain value V decreases by $x\%$, then decreased new value $= (100 - x)\%$ of V

Applications:

Percentage-based problems are commonly seen in real-life situations such as calculating marks, profit or loss, and population growth. They help us understand changes and comparisons between quantities in practical contexts. A good grasp of percentage applications improves accuracy in solving problems related to discounts, interest, and data interpretation. Connecting these problems to daily examples also enhances speed and confidence, especially in competitive examinations.

Population Growth and Decay:

- i. If the population P increases at the rate of $R\%$ per annum, then:

$$\text{Population after years } n \text{ years} = P \left(1 + \frac{R}{100}\right)^n$$

$$\text{Population } n \text{ years ago} = \frac{P}{\left(1 + \frac{R}{100}\right)^n}$$

- ii. If the population P decreases at the rate of $R\%$ per annum, then:

$$\text{Population after years } n \text{ years} = P \left(1 - \frac{R}{100}\right)^n$$

$$\text{Population } n \text{ years ago} = \frac{P}{\left(1 - \frac{R}{100}\right)^n}$$

Depreciation:

If the present value of a machine be P. Suppose it depreciates at the rate of R% per annum. Then:

$$\text{Value of the machine after } n \text{ years} = P \left(1 - \frac{R}{100}\right)^n$$

$$\text{Value of the machine } n \text{ years ago} = \frac{P}{\left(1 - \frac{R}{100}\right)^n}$$

Price of a commodity and expenditure:

If the price of a commodity increases by R%, then the reduction in consumption so as not to increase the expenditure is $\left(\frac{R}{(100 + R)} \times 100\right)\%$

If the price of a commodity decreases by R%, then the increase in consumption so as not to decrease the expenditure is $\left(\frac{R}{(100 - R)} \times 100\right)\%$

SOLVED PROBLEMS

- 3) **Problem 02-03-03:** The price of rice is reduced by 5%. How many kilograms of rice can now be bought for the money which was sufficient to buy 50 kg of rice earlier?

Solution: Let the original price be Rs. 100 per kg.

Money required to buy 50 kg of rice = Rs. (100×50) = Rs. 5000.

New price = Rs. 95 per kg. (Since, price of rice is reduced by 5%)

Quantity of rice can be bought = $\frac{5000}{95} = 52.63 \text{ kg}$.

- 4) **Problem 02-03-04:** A number reduced by 25% becomes 225. What percent should it be increased so that it becomes 390?

Solution: Let the number be x.

It is given that if it is reduced by 25% becomes 225.

$$\Rightarrow (100 - 25)\% \text{ of } x = 225$$

$$\Rightarrow 75 \% \text{ of } x = 225$$

$$\Rightarrow \frac{75}{100} \times x = 225$$

$$\Rightarrow x = 225 \times \frac{100}{75} = 300$$

Suppose, 300 should be increased by $y\%$ so that it becomes 390.

$$\Rightarrow (100 + y)\% \text{ of } 300 = 390$$

$$\Rightarrow \frac{(100 + y)}{100} \times 300 = 390$$

$$\Rightarrow (100 + y) \times 3 = 390$$

$$\Rightarrow (100 + y) = 130$$

$$\Rightarrow y = 30$$

Therefore, given number should be increased by 30% so that it becomes 390.

MCQ

4) **MCQ 02-03-04:** The enrolment of students in a school increases from 480 to 600. What is the percent increase in the enrolment?

- (a) 20% (b) 25%
(c) 30 % (d) 35%

5) **MCQ 02-03-05:** If x is 25% less than y , then what percent is y more than x ?

- (a) 25% (b) 30%
(c) $33\frac{1}{3}\%$ (d) 15%

6) **MCQ 02-03-06:** The population of a town is 8500. It increases by 20% in the first year and by another 25% in the second year. What would be the population of the town after 2 years?

- (a) 10950 (b) 11950
(c) 12550 (d) 12750

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

3) **SAQ 02-03-03:** Madhu's present salary is Rs. 35,000. It will increase by 15% next year. What will be Madhu's salary after the increment?

Solution: Madhu's salary after the 15% increment = $(100 + 15)\%$ of 35000

$$= (115)\% \text{ of } 35000$$

$$= \frac{115}{100} \times 35000$$

$$= 115 \times 350 = \text{Rs. } 40,250.$$

4) SAQ 02-03-04: The population of a town 2 years ago was 62,500. Due to migration to big cities, it decreases every year at the rate of 4%. The find present population of the town.

Solution: If the population P decreases at the rate of $R\%$ per annum, then:

$$\text{Population } n \text{ years ago} = \frac{P}{\left(1 - \frac{R}{100}\right)^n}$$

$$\Rightarrow \text{Present population } P = \text{Population } n \text{ years ago} \times \left(1 - \frac{R}{100}\right)^n$$

From, given information population of a town 2 years ago = 62,500 and

$$n = 2, R = 4.$$

$$\begin{aligned} \therefore \text{Present population } P &= \text{Population 2 years ago} \times \left(1 - \frac{4}{100}\right)^2 \\ &= 62,500 \times \left(1 - \frac{4}{100}\right)^2 = 57,600. \end{aligned}$$

SUMMARY

This unit explains the concept of percentages as a method of expressing numbers as parts of 100 and applies it to various real-life and mathematical situations. It covers basic percentage formulas, conversion between fractions–decimals–percentages, and efficient shortcut methods for quick calculations essential for competitive exams. The unit also explores percentage change, including increase and decrease, successive percentage changes, and comparison-based problems. Students learn practical problem-solving techniques for profit–loss, discounts, population change, marks, mixtures, and data interpretation, enabling them to compute results quickly and accurately.

END OF UNIT EXERCISES

- 1) A batsman scored 110 runs which included 3 boundaries and 8 sixes. What percent of his total score did he make by running between the wickets?
- 2) Three candidates contested an election and received 5100, 5400 and 4500 votes respectively. What percentage of the total votes did the winning candidate get?

- 3) Twenty-five percent of Ravi's yearly income is equal to seventy-five percent of Aakash's monthly income. If Aakash's yearly income is Rs. 360000, what is Ravi's monthly income?
- 4) The marked price of a bicycle is Rs. 8000. Two successive discounts of 10% and $x\%$ are offered. If the customer pays Rs. 6480, determine x .
- 5) The population of a town is 18,54,000. If it had been increasing at 6% per annum, what was its population 5 years ago?

KEY WORDS

Percent, Fraction, Decimal, Percentage Increase, Percentage Decrease

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UNIT 02-04: SEQUENCES AND SERIES

LEARNING OBJECTIVES

After successful completion of this unit, you will be able to

- ❖ Define and distinguish between sequence and series.
- ❖ Identify different types of sequences and series.
- ❖ Understand the concepts of Arithmetic Progression (A.P.), Geometric Progression (G.P.), and Harmonic Progression (H.P.).
- ❖ Derive and apply the formulas for n th term and sum of n terms.
- ❖ Solve real-life and competitive exam-based problems involving progressions.

INTRODUCTION



Arithmetic
Progression
(A.P.)



Geometric
Progression
(G.P.)



Harmonic
Progression
(H.P.)

Patterns and arrangements are an integral part of our daily life and mathematical thinking. Whether it is the arrangement of seats in a classroom, the steps of a staircase, or the regular savings made every month, each of these follows a definite rule or order. Such ordered arrangements of numbers or quantities are known as sequences. A sequence therefore represents a systematic arrangement of terms according to a specific rule that determines how one term leads to the next.

When the terms of a sequence are added together, the resulting expression is known as a series. The study of sequences and series enables us to understand the nature of progression — how quantities grow, decrease, or repeat according to a pattern. These concepts are not merely theoretical but have broad practical applications in fields such as finance, commerce, science, and data analysis.

For example, calculating the interest on a bank deposit, predicting population growth, or determining the depreciation of a machine's value are all real-life

situations where sequences and series are used. In academic studies and competitive examinations, questions based on arithmetic, geometric, and harmonic progressions are commonly asked to assess a learner's logical reasoning and numerical ability.

Thus, understanding the fundamental principles of sequences and series helps learners recognize patterns, establish mathematical relationships, and apply them effectively to both real-world problems and quantitative aptitude tests.

02-04-01: ARITHMETIC PROGRESSION (A.P.)

Sequence: A sequence is an ordered list of numbers following a specific rule or pattern.

Terms of a sequence are denoted by $a_1, a_2, a_3, \dots$ the subscript denotes the position of the term.

A sequence may be finite or infinite, based on how many terms it contains.

Series: If $a_1, a_2, a_3, \dots, a_n, \dots$ is the sequence, then the expression $a_1 + a_2 + a_3 + \dots$ is called the series associated with given sequence. The series is finite or infinite according as the given sequence is finite or infinite.

Progressions: Sequences that follow specific patterns are usually referred to as progressions. In a progression, every term after the first follows a fixed rule or pattern.

Arithmetic progression (A.P.): A sequence is said to be an Arithmetic Progression if the difference between consecutive terms is constant.

This constant difference is called the common difference (d).

General form of A.P. with first term a and common difference d : $a, a + d, a + 2d, a + 3d, \dots$

Note:

- The general term or the n th term of the A.P. is given by $a_n = a + (n - 1)d$.
- If l is the last term of A.P. then n th term from the last $= l - (n - 1)d$.
- The sum S_n of the first n terms of an A.P. is given by
$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l), \text{ where } l = a + (n - 1)d \text{ is the last term of A.P.}$$
- If a, b and c are in A.P., then b is called the Arithmetic Mean (A.M.) of numbers a and c , and $b = \frac{a+c}{2}$.

Illustration: Consider, A.P. 2, 5, 8.

$$\text{Here, } 5 = \frac{2+8}{2}$$

- If the terms of an A.P. are increased, decreased, multiplied or divided by the same constant, they still remain in A.P.

Illustration: Consider, sequence 1, 4, 7, 10, 13, 16, 19, ...

This is A.P. with first term $a = 1$ and common difference $d = 3$

Multiplying each term of the above sequence by 2,

Obtained new sequence: 2, 8, 14, 20, 26, 32, 38, ...

This is also A.P. and first term = 2 and common difference $d = 6$.

- If $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ are two A.P., then
 - i. Addition of A.P. $a_1 + b_1, a_2 + b_2, a_3 + b_3, \dots$ are also in A.P.
 - ii. Multiplication of A.P. $a_1 b_1, a_2 b_2, a_3 b_3, \dots$ and Division of A.P. $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \dots$ are not in A.P.

Illustration: Consider, A.P.s 1, 4, 7, 10, 13, 16, 19, ... and 2, 6, 10, 14, 18, ...

Addition of these A.P.s $1+2, 4+6, 7+10, 10+14, \dots$ that is 3, 10, 17, 24, ... is also A.P. with first term 3 and common difference 7.

But the sequence obtained by the multiplication of these A.P.s $1 \times 2, 4 \times 6, 7 \times 10, 10 \times 14, \dots$ means 2, 24, 70, 140, ... is not A.P.

- If $a_1, a_2, a_3, \dots$ and a_n are in A.P. then
 - i. $a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2} = \dots$

Illustration: Consider, A.P.s 1, 4, 7, 10, 13, 16, 19, 22, 25, 28.

$a_n = a + (n - 1)d$, where $a = 1, d = 3, n = 1, 2, 3, \dots, 10$.

$$a_1 + a_n = a_1 + a_{10} = 1 + 28 = 29$$

$$a_2 + a_{n-1} = a_2 + a_9 = 4 + 25 = 29$$

$$a_3 + a_{n-2} = a_3 + a_8 = 7 + 22 = 29$$

⋮

- ii. If n^{th} term of any sequence is linear expression in n , then the sequence is an A.P.

Illustration: Suppose the n^{th} term is:

$$a_n = 5n + 2$$

This is a linear expression in n .

Let us find the first few terms:

$$a_1 = 5(1) + 2 = 7$$

$$a_2 = 5(2) + 2 = 12$$

$$a_3 = 5(3) + 2 = 17$$

$$a_4 = 5(4) + 2 = 22$$

Now check the differences:

$$a_2 - a_1 = 12 - 7 = 5$$

$$a_3 - a_2 = 17 - 12 = 5$$

$$a_4 - a_3 = 22 - 17 = 5$$

Since the common difference is constant (5), the sequence is an Arithmetic Progression.

General Reason: If $a_n = pn + q$ (linear), then $a_{n+1} - a_n = [p(n+1) + q] - [pn + q] = pn + p + q - pn - q = p$ which is constant $\Rightarrow$ A.P.

- iii. If sum of n terms of any sequence is a quadratic expression in n , then sequence is an A.P.

Illustration: Suppose the sum of n terms is:

$$S_n = 3n^2 + 2n$$

This is a **quadratic expression** in n .

We know:

$$a_n = S_n - S_{n-1}$$

Compute a few terms:

Step 1: Compute S_1, S_2, S_3, S_4 ,

$$S_1 = 3(1)^2 + 2(1) = 5$$

$$S_2 = 3(2)^2 + 2(2) = 16$$

$$S_3 = 3(3)^2 + 2(3) = 33$$

$$S_4 = 3(4)^2 + 2(4) = 56$$

Step 2: Find terms using $a_n = S_n - S_{n-1}$

$$a_1 = S_1 = 5$$

$$a_2 = S_2 - S_1 = 16 - 5 = 11$$

$$a_3 = S_3 - S_2 = 33 - 16 = 17$$

$$a_4 = S_4 - S_3 = 56 - 33 = 23$$

Step 3: Check common difference

$$a_2 - a_1 = 11 - 5 = 6$$

$$a_3 - a_2 = 17 - 11 = 6$$

$$a_4 - a_3 = 23 - 17 = 6$$

Since the difference is constant (6), the sequence is an A.P.

SOLVED PROBLEMS

1) Problem 02-04-01: In an AP, the ratio of the 7th term to the 10th term is 7:10. Find the common difference.

Solution: nth term of the A.P. is given by $a_n = a + (n - 1)d$, where a is the first term and d is common difference and it is given that

$$\frac{a_7}{a_{10}} = \frac{7}{10}$$

$$\Rightarrow \frac{a + 6d}{a + 9d} = \frac{7}{10}$$

$$\Rightarrow 10(a + 6d) = 7(a + 9d)$$

$$\Rightarrow 10a + 60d = 7a + 63d$$

$$\Rightarrow 3a = 3d$$

$$\Rightarrow a = d$$

In an AP, common difference is equal to the first term.

2) Problem 02-04-02: For an A.P. if $S_n = 2n^2 + n$. Find the 10th term.

Solution: We know that, for an A.P. nth term $a_n = S_n - S_{n-1}$.

$$\Rightarrow a_n = [2n^2 + 1n] - [2(n - 1)^2 + (n - 1)]$$

$$\Rightarrow a_n = [2n^2 + 1n] - [2(n^2 - 2n + 1) + (n - 1)]$$

$$\Rightarrow a_n = [2n^2 + 1n] - [2n^2 - 4n + 2 + n - 1]$$

$$\Rightarrow a_n = [2n^2 + 1n] - [2n^2 - 3n + 1]$$

$$\Rightarrow a_n = [2n^2 + 1n] - [2n^2 - 3n + 1]$$

$$\Rightarrow a_n = 4n - 1$$

$$10\text{th term} = a_{10} = 4(10) - 1 = 39.$$

MCQ

- 1) **MCQ 02-04-01:** If 4, x , 16 are the three consecutive terms in A.P., then $x = \dots$
- (a) 6 (b) 12
(c) 14 (d) 10
- 2) **MCQ 02-04-02:** If the sum of first n terms of a sequence is $S_n = 3n^2 + 2n$ then the n th term is:
- (a) $6n+1$ (b) $3n-1$
(c) $6n-1$ (d) $3n+1$
- 3) **MCQ 02-04-03:** In an AP, the 5th term is 10 and the sum of the first 10 terms is 110. Find the 7th term.
- (a) 12 (b) 16
(c) 14 (d) 10
- 4) **MCQ 02-04-04:** If 5th term of an A.P. is 22 and common difference is 4, the first term is:
- (a) 6 (b) 2
(c) 4 (d) 10

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

- 1) **SAQ 02-04-01:** If x, y, z are in A.P., show that the three numbers $x^3 + x^2y + xy^2 + y^3$, $z^3 + z^2x + zx^2 + x^3$, $y^3 + y^2z + yz^2 + z^3$ are consecutive terms of an A.P.

Solution: x, y, z are in A.P. $\Rightarrow y = \frac{x+z}{2}$

Let,

$$T1 = x^3 + x^2y + xy^2 + y^3 = (x^2 + y^2)(x + y)$$

$$T2 = z^3 + z^2x + zx^2 + x^3 = (z^2 + x^2)(z + x)$$

$$T3 = y^3 + y^2z + yz^2 + z^3 = (y^2 + z^2)(y + z)$$

Consider, $T1 + T3 = (x^2 + y^2)(x + y) + (y^2 + z^2)(y + z)$

Using $y = \frac{x+z}{2}$,

$$T1 + T3 = \left(x^2 + \left(\frac{x+z}{2} \right)^2 \right) \left(x + \left(\frac{x+z}{2} \right) \right) + \left(\left(\frac{x+z}{2} \right)^2 + z^2 \right) \left(\left(\frac{x+z}{2} \right) + z \right)$$

$$T1 + T3 = 2(z^2 + x^2)(z + x) = 2T2$$

$$\Rightarrow T2 = \frac{T1 + T3}{2}$$

$T1, T2, T3$ are consecutive terms of an A.P.

2) SAQ 02-04-02: The product of three numbers in A.P. is 120, and the largest number is 15 times the smallest. Find the numbers.

Solution: Let the three numbers in A.P. be $a - d, a, a + d$ ($d > 0$).

Product of three numbers in A.P. is 120 $\Rightarrow (a - d) \times a \times (a + d) = 120$

$$\Rightarrow a(a^2 - d^2) = 120 \dots \dots \dots (1)$$

Further it is given that, the largest number is 15 times the smallest.

$$\Rightarrow a + d = 15(a - d)$$

$$\Rightarrow a + d = 15(a - d)$$

$$\Rightarrow 16d = 14a$$

$$\Rightarrow d = \frac{7}{8}a$$

Therefore, substituting $d = \frac{7}{8}a$ in equation (1) we get,

$$a(a^2 - \left(\frac{7}{8}a\right)^2) = 120$$

$$\Rightarrow a(a^2 - \frac{49}{64}a^2) = 120$$

$$\frac{15a^3}{64} = 120$$

$$a^3 = 120 \times \frac{64}{15}$$

$$\Rightarrow a = 8$$

$$\Rightarrow d = \frac{7}{8}a = 7$$

Hence, three numbers in A.P. are 1, 8, 15.

02-04-02: GEOMETRIC PROGRESSION (G.P.) AND HARMONIC PROGRESSION (H.P.)

Geometric progression (G.P.): A sequence is said to be a Geometric Progression if the ratio of consecutive terms is constant. This constant is called the common ratio (r).

In Geometric Progression each term except the first is obtained by multiplying the previous term by a non-zero constant called the common ratio.

General form of G.P. with first term a and common ratio r : $a, ar, ar^2, ar^3, \dots$

Note:

- The general term or the n^{th} term of the G.P. is given by $a_n = ar^{n-1}$.
- If l is the last term of G.P. then n^{th} term from the last $a_n = \frac{l}{r^{n-1}}$.
- The sum S_n of the first n terms of a G.P. is given by

$$S_n = \frac{a(r^n - 1)}{(r - 1)} \quad \text{if } r \neq 1$$

$$S_n = na \quad \text{if } r = 1$$

- If a, b and c are in G.P., then b is called the Geometric Mean (G.M.) of numbers a and c , and it is given by $b = \sqrt{ac}$.

Illustration: Consider, G.P. 2, 4, 8.

Here, $4 = \sqrt{2 \times 8}$

- If the terms of a G.P. are multiplied or divided by the same non-zero constant, they still remain in G.P.

Illustration: Consider, sequence 2, 4, 8, 16, ...

This is G.P. with first term $a = 2$ and common ratio $r = 2$

Dividing each term of the above sequence by 2,

Obtained new sequence: 1, 2, 4, 8, ...

This is also G.P. and first term = 1 and common ratio = 2.

- If $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ are two G.P.s, then Multiplication of G.P.s $a_1 b_1, a_2 b_2, a_3 b_3, \dots$ and Division of G.P.s $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \dots$ are also in G.P.

Illustration: Consider, G.P.s 1, 2, 4, 8, 16, ... and 3, 9, 27, 81, 243, ...

Sequence obtained by the multiplication of these G.P.s $1 \times 3, 2 \times 9, 4 \times 27, 8 \times 81, \dots$ means 3, 18, 108, 648, ... is also G.P. with first term 3 and common ratio 6.

- If $a_1, a_2, a_3, \dots$ are in G.P. then $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots$ are also in G.P.

Illustration: Consider, G.P. 1, 2, 4, 8, 16, ... with the first term = 1 and common ratio = 2.

Sequence of the reciprocals of terms in this G.P. $\frac{1}{1}, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$ is also G.P. with first term 1 and common ratio $\frac{1}{2}$.

- If $a_1, a_2, a_3, \dots$ are in A.P. ($a_i > 0, \forall i$), then $x^{a_1}, x^{a_2}, x^{a_3}, \dots$ are in G.P. ($\forall x > 0$).

Illustration: Consider, A.P. 1, 3, 5, 7, 9, ... and $x = 2$, then $x^{a_1}, x^{a_2}, x^{a_3}, \dots$

Means $2^1, 2^3, 2^5, 2^7, \dots$ are in G.P. with common ratio 2^2 .

- If $a_1, a_2, a_3, \dots$ and a_n are in G.P. then $a_1 a_n = a_2 a_{n-1} = a_3 a_{n-2} = \dots$

Illustration: Consider, G.P.s 1, 2, 4, 8, 16, ..., 512.

Here, $a_n = ar^{n-1}$, with $a = 1, r = 2, n = 1, 2, 3, \dots, 10$.

$$a_1 a_n = a_1 a_{10} = 1 \times 2^9 = 2^9$$

$$a_2 a_{n-1} = a_2 a_9 = 2 \times 2^8 = 2^9$$

$$a_3 a_{n-2} = a_3 a_8 = 4 \times 2^7 = 2^9.$$

Harmonic Progression (H.P.): A sequence $a_1, a_2, a_3, \dots$ is in Harmonic Progression if the reciprocals $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots$ form an Arithmetic Progression (A.P.).

General Term of H.P.: If $a_1, a_2, a_3, \dots$ are in H.P. Let $\frac{1}{a_n} = A + (n-1)d$, where $A = \frac{1}{a_1}$ and d is the common difference of

the A.P. $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots$

Then n^{th} term of H.P. is given by

$$a_n = \frac{1}{A + (n-1)d}$$

Note:

- If a, b and c are in H.P., then b is called the **Harmonic Mean (H.M.)** of numbers a and c , and it is given by $b = \frac{2ac}{a+c}$.

- **Relation Between A.M., G.M., H.M.:** For two positive numbers:

$$A.M. \geq G.M. \geq H.M.$$

Important Results on Sum of Special Sequences:

1. Sum of First n Natural Numbers

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

2. Sum of Squares First n Natural Numbers

$$1 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

3. Sum of Cubes First n Natural Numbers

$$1 + 2^3 + 3^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

4. Sum of First n Odd Numbers

$$1 + 3 + 5 + \cdots + (2n-1) = n^2$$

5. Sum of First n Even Numbers

$$2 + 4 + 6 + \cdots + 2n = n(n+1)$$

SOLVED PROBLEMS

- 3) **Problem 02-04-03:** Find the sum of the first 6 terms of the GP with $a = 5$ and common ratio $r = 2$.

Solution: The sum S_n of the first n terms of a G.P. is given by

$$S_n = \frac{a(r^n - 1)}{(r - 1)}$$

Therefore, sum of the first 6 terms of the GP

$$S_6 = \frac{a(r^6 - 1)}{(r - 1)} = \frac{5(2^6 - 1)}{(2 - 1)} = 5 \times 63 = 315$$

- 4) **Problem 02-04-04:** If 2, x, 18 are in GP, find x.

Solution: If a, b and c are in G.P., then b is called the Geometric Mean (G.M.) of numbers a and c, and it is given by $b = \sqrt{ac}$.

Here 2, x, 18 are in GP $\Rightarrow x = \sqrt{2 \times 18} = \sqrt{36} = 6$.

MCQ

- 5) **MCQ 02-04-05:** The third term of G.P. is 6. The product of its first 5 terms is

(a) 30 (b) 6^4

(c) 6^5 (d) 6^3

6) **MCQ 02-04-06:** $2 + 4 + 6 + \dots + 200 =$

(a) 10100 (b) 5050

(c) $(5050)^2$ (d) $(10100)^2$

7) **MCQ 02-04-07:** If the first term of a GP is 3 and the 4th term is 24, the common ratio is:

(a) 2 (b) 4

(c) 8 (d) 3

8) **MCQ 02-04-08:** Harmonic Mean of 2 and 3 is

(a) $\frac{5}{2}$ (b) $\frac{12}{5}$

(c) $\sqrt{6}$ (d) 5

SHORT ANSWER QUESTIONS WITH MODEL ANSWER

3) **SAQ 02-04-03:** A car depreciates in value at the rate of 10% per year, calculated on its value at the beginning of each year. If the initial price of the car was ₹50,000, find its value at the end of 3 years.

Solution: Depreciation rate = 10% per year
Remaining value each year = $100\% - 10\% = 90\% = 0.90$

This forms a Geometric Progression (GP) with:

Initial value $a = ₹ 50,000$

Common ratio $r = 0.90$

Number of years $n = 3$

Value of car after 3 years = $a \times r^n = 50,000 \times (0.9)^3 = 50,000 \times 0.729 = 36,450$

4) **SAQ 02-04-04:** If 2, 3, x are in Harmonic Progression, find x.

Solution: 2, 3, x are in Harmonic Progression

$\Rightarrow \frac{1}{2}, \frac{1}{3}$ and $\frac{1}{x}$ are in A.P.

$$\Rightarrow \frac{1}{3} - \frac{1}{2} = \frac{1}{x} - \frac{1}{3}$$

$$\Rightarrow \frac{1}{3} - \frac{1}{2} = \frac{1}{x} - \frac{1}{3}$$

$$\Rightarrow -\frac{1}{6} = \frac{1}{x} - \frac{1}{3}$$

$$\Rightarrow \frac{1}{3} - \frac{1}{6} = \frac{1}{x}$$

$$\Rightarrow \frac{1}{x} = \frac{1}{6}$$

$$\Rightarrow x = 6.$$

SUMMARY

This unit introduces the foundational concepts of sequences and series, focusing on three important progressions used in mathematics and competitive examinations. A sequence is an ordered list of numbers following a specific pattern, while a series is the sum of the terms of a sequence. Different types of sequences—finite, infinite, increasing, decreasing, periodic, and special patterns—help describe numerical behavior in various mathematical and real-life applications.

The unit covers Arithmetic Progression (A.P.), where each term increases or decreases by a constant difference; Geometric Progression (G.P.), in which each term is multiplied by a constant ratio; and Harmonic Progression (H.P.), defined through the reciprocals of an A.P. Key formulas for the general term and sum of A.P. and G.P. enable efficient computation, while properties of H.P. are derived using reciprocals. The unit also includes special sequences, such as sums of the first n natural numbers, squares, and cubes, often used in competitive exam shortcuts.

END OF UNIT EXERCISES

- 1) A carpenter was hired to build 200 window frames. The first day he made four frames and each day, thereafter he made two more frames than he made the day before. How many days did it take him to finish the job?
- 2) Calculate: $1 + 4 + 9 + \dots + 289$

- 3) In a G.P., the 2nd term is 6 and the 5th term is 162. Find the common ratio.
- 4) Find the sum of first 10 terms of the GP: 2, 6, 18, 54, ...
- 5) Find the tenth term of the Harmonic Progression: $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots$

KEY WORDS

Sequences, Series, Arithmetic Progression, Geometric Progression, Harmonic Progression

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ANSWERS TO MCQ AND HINTS TO END OF UNIT EXERCISES

Unit 01-01: Numbers

MCQ

MCQ 01-01-01: (d); MCQ 01-01-02: (a); MCQ 01-01-03: (b);
MCQ 01-01-04: (c); MCQ 01-01-05: (a); MCQ 01-01-06: (b);
MCQ 01-01-07: (b); MCQ 01-01-08: (b); MCQ 01-01-09: (d);
MCQ 01-01-10: (a).

Hints to End of Unit Exercises

- 1) Divisibility by 99 (114345, 813464)
Use: $99 = 9 \times 11$
Check divisibility by 9 (sum of digits) and 11 (alternate sum rule).
- 2) Calculate $1 + 2^3 + 3^3 + \dots + 50^3$
Use formula: Sum of cubes = $\left(\frac{n(n+1)}{2}\right)^2$
Put, $n = 50$.
- 3) Find digit * in 78*3945 divisible by 11
Apply divisibility rule of 11:
(Sum of odd place digits – sum of even place digits) must be multiple of 11.
- 4) 476**0 divisible by both 3 and 11
Two conditions:
For 3: Sum of digits divisible by 3
For 11: Difference of alternate sums divisible by 11
Solve simultaneously.
- 5) Sum of all primes from 1 to 20
List primes ≤ 20 : 2, 3, 5, 7, 11, 13, 17, 19
Then add them.

Unit 01-02: H.C.F. and L.C.M. of Numbers

MCQ

MCQ 01-02-01: (a); MCQ 01-02-02: (c); MCQ 01-02-03: (d);
MCQ 01-02-04: (d); MCQ 01-02-05: (c); MCQ 01-02-06: (b);

MCQ 01-02-07: (d); MCQ 01-02-08: (a); MCQ 01-02-09: (b);
MCQ 01-02-10: (d); MCQ 01-02-11: (b); MCQ 01-02-12: (b).

Hints to End of Unit Exercises

- 1) H.C.F. of 884, 1210, and 1596

Use the division (Euclidean) method stepwise:

- First find HCF of 1596 and 1210
- Then take that result and find HCF with 884

- 2) H.C.F. of 72 and 108 (Prime Factorization Method)

Factorize both numbers into primes

Take product of common prime factors with the smallest common powers

- 3) H.C.F. of 98 and 63 (Division Method)

Apply Euclidean algorithm:

- Divide the larger number by the smaller
- Continue until remainder becomes 0
- The last divisor is the HCF

- 4) Product = 1500, H.C.F. = 10 → Find L.C.M.

Use formula: $L.C.M. \times H.C.F. = \text{Product of numbers}$

- 5) L.C.M. of 14, 28, 35 (Prime Factorization Method)

Factorize numbers into primes

Take product of highest powers of all distinct prime factors

- 6) L.C.M. of 9, 12, 15 (Common Division Method)

Arrange the numbers and divide by a common divisor for at least two until no numbers share a divisor except 1. Then, calculate the L.C.M. by multiplying all divisors and undivided numbers.

- 7) H.C.F. of 1.75, 5.6, and 7

First convert decimals into whole numbers multiplying by 100

Find HCF of resulting whole numbers 175, 560, 700.

Divide obtained HCF by 100

- 8) $L.C.M. \text{ of fractions} = \frac{\text{L.C.M. of Numerators}}{\text{H.C.F. of Denominators}}$

Unit 01-03: Square Root and Cube Root

MCQ

MCQ 01-03-01: (b); MCQ 01-03-02: (c); MCQ 01-03-03: (b);
MCQ 01-03-04: (b); MCQ 01-03-05: (d); MCQ 01-03-06: (b);
MCQ 01-03-07: (c).

Hints to End of Unit Exercises

- 1) Start from the innermost square root $\sqrt{144} = 12$, then substitute step-by-step $\sqrt{64}$ and $\sqrt{256}$.
- 2) 79507
Last digit = 7 $\rightarrow$ cube root ends with 3
Removing last 3 digits:
79507 $\rightarrow$ 79
Nearest cube ≤ 79 is $4^3 = 64$.
- 3) $\sqrt{441} = 21$
- 4) $\sqrt{64} = 8$ and $\sqrt{9} = 3$
- 5) $\sqrt{81} = 9, \sqrt{121} = 11, \sqrt{16} = 4$
- 6) Let, $x = \sqrt{6+x}$, Square and factories

Unit 01-04: Simplification

MCQ

MCQ 01-04-01: (c); MCQ 01-04-02: (b); MCQ 01-04-03: (c);
MCQ 01-04-04: (c); MCQ 01-04-05: (b); MCQ 01-04-06: (a);
MCQ 01-04-07: (c); MCQ 01-04-08: (c); MCQ 01-04-09: (b);
MCQ 01-04-10: (c); MCQ 01-04-11: (c); MCQ 01-04-12: (a);
MCQ 01-04-13: (b).

Hints to End of Unit Exercises

- 1) First calculate: $3565 \div 23$
Subtract this value from 430, then solve:
 $4675 \div ? = \text{remaining value}$
- 2) Use BODMAS
First calculate: 36 of $\frac{1}{6}$

Then perform division and multiplication separately.

- 3) Solve the bracket first: 5×3

Then divide 45 by the result and add 4.

- 4) First multiply: 2×3

Then solve inside the bracket and subtract from 20.

- 5) Use order of operations:

I. Exponents

II. Division and multiplication from left to right

III. Addition/subtraction

Also, $a \div \frac{1}{3} = a \times 3$

- 6) Let the number be x .

Correct product: $52x$ and Wrong product: $25x$

Difference: $52x - 25x = 945$. Solve for x .

Unit 02-01: Ratios and Proportions

MCQ

MCQ 02-01-01: (c); MCQ 02-01-02: (d); MCQ 02-01-03: (c);

MCQ 02-01-04: (d); MCQ 02-01-05: (c); MCQ 02-01-06: (d);

MCQ 02-01-07: (b); MCQ 02-01-08: (b); MCQ 02-01-09: (d).

Hints to End of Unit Exercises

- 1) Find original shares using ratio 3: 2:

$$\text{Share} = \frac{\text{part}}{\text{total parts}} \times 25000$$

Add Rs. 5000 to each share and form the new ratio.

- 2) Given: 1 star = 4 circles and 3 circles = 4 diamonds

Convert both into circles (or diamonds) and compare.

- 3) Convert both quantities to the same unit first: Rs. 4 = 400 paise

Then simplify the ratio.

- 4) Raj drank: $\frac{1}{3} \times 3$ litres

Subtract from total water to get Suraj's share.

- 5) Convert both times into the same unit: 7 hours 30 minutes = 7.5 hours

or convert both into minutes, then simplify the ratio.

Unit 02-02: Logarithms

MCQ

MCQ 02-02-01: (b); MCQ 02-02-02: (b); MCQ 02-02-03: (b);
MCQ 02-02-04: (b); MCQ 02-02-05: (c); MCQ 02-02-06: (d);
MCQ 02-02-07: (a).

Hints to End of Unit Exercises

- 1) Write all numbers as powers of 5: $125 = 5^3, 625 = 5^4, 25 = 5^2$
- 2) Use logarithm laws: $\log a - \log b + \log c = \log\left(\frac{ac}{b}\right)$
- 3) Simplify using $\log_5 5 = 1, \log_4 9 = \frac{\log 9}{\log 4} = \frac{\log 3^2}{\log 2^2}$
- 4) Simplify from inside outward: $27^3 = (3^3)^3 = 3^9$
- 5) Express each number as a power of its base.

Unit 02-03: Percentages

MCQ

MCQ 02-03-01: (d); MCQ 02-03-02: (b); MCQ 02-03-03: (b);
MCQ 02-03-04: (b); MCQ 02-03-05: (c); MCQ 02-03-06: (d).

Hints to End of Unit Exercises

- 1) Runs from boundaries and sixes: $3 \times 4 + 8 \times 6$
Subtract from 110 to get runs by running. Then find:

$$\frac{\text{runs by running}}{110} \times 100$$

- 2) Total votes: $5100 + 5400 + 4500$

Winning candidate got 5400 votes.

$$\text{Find: } \frac{5400}{\text{total votes}} \times 100$$

- 3) Aakash's monthly income: $\frac{360000}{12}$

25% of Ravi's yearly income = 75% of Aakash's monthly income

Find Ravi's yearly income, then divide by 12.

4) After 10% discount Price = $8000 \times \frac{90}{100}$

Then apply $x\%$ discount and equate final price to 6480.

5) Use population growth formula for $P = 18,54,000$, $R = 6\%$ p.a.

$$\text{Population } n \text{ years ago} = \frac{P}{\left(1 - \frac{R}{100}\right)^n}$$

Unit 02-04: Sequences and Series

MCQ

MCQ 02-04-01: (d); MCQ 02-04-02: (c); MCQ 02-04-03: (c);

MCQ 02-04-04: (a); MCQ 02-04-05: (c); MCQ 02-04-06: (a);

MCQ 02-04-07: (a); MCQ 02-04-08: (b).

Hints to End of Unit Exercises

1) Window Frames (A.P.)

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

with $a = 4$, $d = 2$, $S_n = 200$. Solve for n .

2) $1 + 4 + 9 + \dots + 289 = 1^2 + 2^2 + \dots + 17^2$

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n + 1)(2n + 1)}{6}$$

with $n = 17$.

3) G.P.

$$a_n = ar^{n-1}$$

$$ar = 6, \quad ar^4 = 162$$

Divide the equations to find r .

4) Sum of G.P.

$$S_n = a \left(\frac{r^n - 1}{r - 1} \right)$$

with $a = 2$, $r = 3$, $n = 10$.

5) H.P.

Reciprocals form the A.P.:

$$1, 3, 5, 7, \dots$$

Find the 10th A.P. term using $a_n = a + (n - 1)d$ then take its reciprocal.

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FEEDBACK SHEET FOR THE STUDENT

Dear Student,

You have gone through this book, it is time for you to do some thinking for us. Please answer the following questions sincerely. Your response will help us to analyse our performance and make the future editions of this book more useful. Your response will be completely confidential and will in no way affect your examination results. Your suggestions will receive prompt attention from us.

Please submit your feedback online at this QR Code or at following link



<https://forms.gle/rpDib9sy5b8JEisQ9>

or email at: director.ast@ycmou.ac.in

or send this filled “Feedback Sheet” by post to above address.

(Please tick the appropriate box)

Pl. write your Program Code: Course Code & Name:

Style

01. Do you feel that this book enables you to learn the subject independently without any help from others?

☐	☐	☐
--------------------------	--------------------------	--------------------------

Yes No Not Sure

02. Do you feel the following sections in this book serve their purpose? Write the appropriate code in the boxes.

Code 1 for “Serve the purpose fully”

Code 2 for “Serve the purpose partially”

Code 3 for “Do not serve any purpose”

Code 4 for “Purpose is not clear”

Warming up	☐	Check Point	☐	Answer to Check Points	☐
To Begin With	☐	Summary	☐	References	☐

Objectives

Key Words

03. Do you feel the following sections or features, if included, will enhance self-learning and reduce help from others?

	Yes	No	Not Sure
Index			
Glossary			
List of “Important Terms Introduced”			
Two Colour Printing			

Content

04. How will you rate your understanding of the contents of this Book?

Very Bad	Bad	Average	Good	Excellent

05. How will you rate the language used in this Book?

Very Simple	Simple	Average	Complicated	Extremely Complicated

06. Whether the Syllabus and content of book complement to each other?

Yes	No	Not Sure

07. Which Topics you find most easy to understand in this book?

Sr. No.	Topic Name	Page No.

08. Which Topics you find most difficult to understand in this book?

Sr. No.	Topic Name	Page No.

09. List the difficult topics you encountered in this Book. Also try to suggest how they can be improved.

Use the following codes:

Code 1 for “Simplify Text”

Code 2 for “Add Illustrative Figures”

Code 3 for “Provide Audio-Vision (Audio Cassettes with companion Book)”

Code 4 for “Special emphasis on this topic in counseling”

Sr. No.	Topic Name	Page No.	Required Action Code

10. List the errors which you might have encountered in this book.

Sr. No.	Page No.	Line No.	Errors	Possible Corrections

11. Based on your experience, how would you place the components of distance learning for their effectiveness?

Use the following code.

Code 1 for “Most Effective” Code 3 for “Average” Code 5 for “Least Effective”

Code 2 for “Effective” Code 4 for “less Effective”

Printed Book		Counseling		Lab Journal	
Audio Lectures		Home Assignment		YouTube Videos	
Video Lectures		Lab-Experiment		Online Counseling	

12. Give your overall rating to this book?

1. 2. 3. 4. 5.

13. Any additional suggestions:

Thank you for your co-operation!